

# Predictive Interaction Realizability on the FR3

## From a design gate to a two-axis certificate — what holds, what broke, and what it costs

*Working draft, simulation only. It began as one question — is the passive-nominal split feasible? — and the answer to that (§1–§6) turned out to be the least interesting thing in it. The merged controller (§7), the fixes for what it exposed (§8), the pose study that dissolved most of those (§9), and the tension between PIR’s two axes (§10) each changed the previous answer.*

**How to read this.** The document is written in the order the work happened, and several conclusions were later overturned by a subsequent experiment. Rather than silently rewriting them, superseded sections carry a **Corrected by §x** banner at the top and the correction is made where the newer evidence is. §11.1 tabulates every such reversal, because the pattern in them — five of seven were properties of one FR3 configuration — is itself one of the findings. If you want only the current state: §1, then §10, then §12.

## 1. Decision summary

Merging `phri2`’s behaviour-realization separation with `impedance_residual`’s energy-authorized residual port requires `phri2` to acquire something it does not have: a **nominal**. The proposed route is the *passive-nominal split*,

$$F_{\text{cmd}} = F_{\text{nom}} + F_r, \quad F_{\text{nom}} = -K_0 e - D_0 v,$$

in which  $F_{\text{nom}}$  is a small behaviour-agnostic spring-damper that supplies a passivity floor and a harvestable dissipation term  $v^\top D_0 v$ , and the realization QP optimizes only the residual  $F_r$  that the tank then authorizes. The merged Lemma 1 that this route buys is worthless unless its first precondition is satisfiable: the **anchor**

$$a_\ell = \tau_{\text{base},\ell} + J_{v,\ell}^\top F_{\text{nom},\ell}, \quad |a_\ell| \leq \bar{\tau} \quad \text{at every tick,}$$

must fit inside the derated torque envelope — and the  $\alpha \rightarrow 0$  fallback it degenerates to must still hold a sustained 20 N push inside the 0.06 m workspace bound. Those two requirements pull in opposite directions. This document reports whether they have a common solution.

**They do, but only under one of the two envelopes the source papers use, and only in a sliver.**

| Torque envelope                       | Source                          | Feasible region                                                            | Verdict                               |
|---------------------------------------|---------------------------------|----------------------------------------------------------------------------|---------------------------------------|
| $\rho \tau_{\text{max}}, \rho = 0.28$ | <code>impedance_residual</code> | <b>0 / 900 cells</b>                                                       | <b>NO-GO</b> (at this pose; see §9.2) |
| 87/87/87/ <b>31.5</b> /12/12/12 N·m   | <code>phri2</code>              | <b>33 / 900 cells</b> , $k \in [360, 400]$ N/m, $d \in [21.3, 48.4]$ N·s/m | <b>GO, thin</b>                       |

Under the  $\rho = 0.28$  envelope no  $(K_0, D_0)$  works *at this pose* — and §9.2 later shows that “at this pose” is the whole of it, correcting §5.1’s original claim that the result was structural. Under `phri2`’s own envelope the split is viable at  $K_0 = 380$  N/m,  $D_0 = 29.1$  N·s/m, with every hard diagnostic sitting at 93–98 % of its limit (§5.2).

Three findings change what the merge can claim, and all three are load-bearing:

- The two papers’ “derated budget” stress cases are not interchangeable.** The synthesis note treats both as  $\rho = 0.28$ . They are not: `phri2`’s own realized torque peaks at **1.30×** the  $\rho = 0.28$  cap, so `phri2`’s published FR3 benchmark does not even *run* inside `impedance_residual`’s envelope. The merge inherits whichever envelope it is written against, and the answer flips between them.
- The note’s four-diagnostic table admits false green cells.** Its row (1) is evaluated on a trajectory where  $F_{\text{nom}}$  is small, while its row (3) defines a displaced operating point the robot must actually hold. Checking each in its own place, 70 cells pass; requiring the precondition to hold on the *fallback itself* (row 1b, §4.3) leaves 33. Half the apparent region is an artifact of never asking one trajectory to satisfy both constraints.

3. **The  $\alpha \rightarrow 0$  fallback transiently leaves the workspace box.** At the recommended cell the settled displacement is 55.8 mm against the 60 mm bound, but the transient peaks at **62.2 mm**. (*Originally this finding continued: gate on peak rather than settled displacement and the region drops from 33 cells to 6. §12.0 withdraws that reading — the box is slack-relaxed by necessity in the QP this document reuses, so the 33-cell region stands. What survives is the weaker statement: the  $\alpha \rightarrow 0$  guarantee is “settles inside the box”, not “stays inside it”.*)

The merged controller has since been implemented and run in closed loop (§7). That settles the gate’s main caveat and adds a fourth finding that outranks the other three:

4. **The split converts a hard actuator guarantee into a conditional one.** In `phri2` the QP’s torque constraint covers the *whole* command. Under the split the nominal sits outside the authorization loop — the 1 kHz servo can scale  $F_r$  but never  $F_{\text{nom}}$  — so once  $|\tau_{\text{base}} + J_v^\top F_{\text{nom}}| > \bar{\tau}$  nothing in the fast loop can prevent the overrun. With only 2.4 % of joint 4’s cap left as headroom, that happens at  $8 \times \text{impedance\_residual}$ ’s own disturbance amplitude. (§8.2’s fix, adopted since, removes it: the shipped default holds the envelope at every scale tested. §7.7 records the behaviour that motivated the fix.)

§8 root-causes all four and reports what can be done about them: findings 1–3 are one problem, not three — diagnostic 3 forces  $K_0$  above the desired impedance’s own stiffness — and finding 4 is separate and is fixed outright by giving the servo authority over the nominal (**adopted**, §8.2). §8.5 then states the repaired guarantee, **Merged Lemma 1’**, whose precondition is merely  $|\tau_{\text{base}}| \leq \bar{\tau}$  — a pose property, not a design one — and checks it clause by clause. It also prices the fix — and the price turned out to be the *rule*, not the fix. Forbidding  $\alpha_{\text{nom}}$  to rise made it a latch that spent 71 points of workspace excursion for a 0.74 % overrun of a *derated* cap; **metering the rise against the tank instead** (§8.2, adopted) keeps both guarantees and returns all 71 points at  $8 \times$ . §8.6 then proves what can be proved about recursive feasibility — a sound feasibility certificate, which turns out to be the Task 1 gate’s own row 1b, and an invariant terminal set that is nonempty exactly when the precondition holds strictly. Appendix A collects the merged statements at submittable granularity.

**§9 then removes most of the problem rather than trading it.** All four findings were measured at one FR3 pose, and that pose is a bad one: its bias torque uses 63.6 % of joint 4’s cap before any control is applied. At a pose that does not load joint 4 the feasible region grows from 33 cells to **481**, the residual’s torque headroom from 2.4 % to **45.7 %**, and the workspace excursion under a  $16 \times$  disturbance *improves* from 164 mm to 59 mm. Findings 1–3 were substantially an artifact of the scenario, not of the architecture.

**§10 is the one that changes what the paper claims.** Aggregating every run, the two axes turn out to be *anti-correlated*: at `phri2`’s pose the feasibility axis fires and  $r_{\text{auth}}$  is exactly zero; at the recommended pose  $r_{\text{auth}}$  removes 46 % of the intended behaviour and  $\alpha_r$  never fires at all. Both act on the same residual through the same torque budget, so a nominal stiff enough to make saturation bind leaves too small a residual to drain the tank, and vice versa. The framing that survives this is stronger than “both axes matter”: **you cannot tell in advance which axis will bind**, and one innocuous-looking joint-angle change flips it. A joint pose  $\times K_0$  scan (§10.5, 96 cells) then finds exactly one operating point where both axes carry real load with every certificate intact — and §10.6 resampling it across 10 seeds shows that point keeps all its certificates in only **5 of 20** runs, always failing on Lemma 1’s precondition. The flip, by contrast, is **20/20**. So the claim the evidence supports is the flip, and the dual-axis regime is a remark rather than an exhibit.

**§11 then does two things that narrow the claim.** §11.0 stops disclaiming the force-misclassification pillar and measures it: label a fraction of the human’s force as disturbance and *every* certificate still holds, the four-term closure still closes to machine precision — against the wrong target — and the reported realization residual **falls by a factor of 19 while the true behavioural error more than doubles**. Misclassification does not merely evade the certificates; it registers as an improvement on the primary number §7–§10 report. §11.2 then applies this document’s own rule and sweeps every remaining claim across the pose family: sixteen claims, split exactly in half, with a clean pattern — **the guarantees are pose-invariant, the hypotheses are not**. The torque envelope and the tank floor hold at every pose at every load; the precondition, the headroom and §8.6’s certificate all fail at `phri2`’s end and recover by  $\lambda \approx 0.4$ .

So the honest form of the claim is narrower than it first looks: **PIR certifies that the *declared* behaviour was realized within stated bounds, at poses where a run-time hypothesis it reports but does not defend happens to hold**. Whether the declaration matches what the human is doing is upstream of everything here.

§12 states what is left for a human — which, after §12.0 answered the two specification questions from the sources, is decision 1 and ratifying decision 3.

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## 2. Why the split is needed at all

The two unsubmitted drafts are complementary halves of “predictive impedance under saturation”:

- `impedance_residual` holds a fixed physical impedance as nominal, lets a 100 Hz MPC propose an additive residual wrench, and authorizes that residual at 1 kHz against both a joint-torque envelope and an energy tank. Its central result is that the residual port must be re-authorized *between* manager updates.
- `phri2` emits a desired contact-port acceleration  $a_{id} = C_\theta x + G_\theta F_h$  from a behaviour layer, realizes it in a receding-horizon QP under a horizon-wide torque constraint, and reports the deviation as  $r = r_{\text{reg}} + r_{\text{con}} + r_{\text{mod}}$  rather than hiding it in clipping.

`phri2` §VIII names its own gap in as many words: “the realization cost does not itself imply passivity ... A dissipativity constraint on the realized port variables, paired with an energy tank or passivity observer, is a natural extension.” That extension is `impedance_residual`’s tank.

The graft does not work naively, for a structural reason. `impedance_residual`’s tank sits on a nominal  $\tau_I$  that is always-feasible and natively passive, and it authorizes only the additive residual. `phri2` **has no nominal**. Its  $\tau_{\text{base}}$  is feedforward plus orientation hold plus null-space centering, with no impedance term at all; the entire interaction behaviour lives inside  $F_{\text{cmd}}$ . Two consequences follow:

1. Scaling  $F_{\text{cmd}}$  down at a fast tick degenerates not to a passive spring-damper but to **zero interaction behaviour** — gravity compensation and posture only, a free-floating arm.
2. The tank’s measured advantage comes from harvesting the nominal damping  $v^\top D_I v$ . `phri2` has no harvestable damping term.

The passive-nominal split exists to fix both at once, at the cost of introducing  $(K_0, D_0)$  — and with them the question this document answers.

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## 3. The merged fast re-authorization rule

For context, the rule the split is meant to enable. The QP supplies  $F_{r,k}$  every 20 ms; the following runs every 1 kHz tick  $\ell$  ( $h = 1$  ms):

1. Recompute from the current  $(q, \dot{q})$ :  $\tau_{\text{base},\ell}$ ,  $J_{v,\ell}$ ,  $\Lambda_\ell$ , and  $F_{\text{nom},\ell} = -K_0 e_\ell - D_0 v_\ell$ .
2. **Anchor**  $a_\ell = \tau_{\text{base},\ell} + J_{v,\ell}^\top F_{\text{nom},\ell}$ , with the precondition  $|a_\ell| \leq \bar{\tau}$ .
3. **Torque scale**  $\alpha_{\tau,\ell} = \text{largest } \alpha \in [0, 1] \text{ with } |a_\ell + \alpha J_{v,\ell}^\top F_{r,k}| \leq \bar{\tau} \text{ per joint.}$
4. **Energy scale**  $\alpha_{E,\ell} = 1$  if  $\bar{F}_{r,\ell}^\top v_\ell \leq 0$ , else  $\min(1, (E_\ell - E_{\min} + h v_\ell^\top D_0 v_\ell) / (h \bar{F}_{r,\ell}^\top v_\ell))$ .
5. **Apply** and **ledger** as in `impedance_residual`.

Steps 1-2 are what §4-§6 gate. Steps 3-6 are implemented in `pir_controller.py` and measured in §7.

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## 4. Method

### 4.1 One FR3 model, imported not rewritten

Every mass matrix, Jacobian, task-space inertia, bias torque, null-space projector, torque limit and scenario constant is imported from the existing studies ( `simulation/fr3_mujoco.py` , `simulation/fr3_impedance.py` , `imp_reference/simulation/fr3_interaction_dynamics_mpc.py` , `imp_reference/simulation/run_fr3_experiments.py` ). Nothing is re-derived: a second FR3 model would make this gate inconsistent with the papers it gates. The regression suite asserts the 0.06 m bound *is* `FR3MPCConfig.position_limit` rather than merely equalling it.

### 4.2 Scenario

`phri2`’s FR3 benchmark: the EE holds a fixed nominal pose while a smooth sustained 20 N push along  $-z$  (raised-cosine ramp, `human_force_at` ) would, under naive realization, drive it through the 0.06 m workspace bound. 6 s, 1 kHz servo, 50 Hz manager. Task-space inertia at the nominal pose is  $\Lambda_{zz} = 4.56$  kg; the bias torque on joint 4 is 18.96 N·m before any control is applied at all.

Two torque envelopes are carried, because the two papers use different ones:

- **rho\_0.28** —  $0.28 \tau_{\max}$  on every joint (24.36 N·m on joints 1–4, 3.36 N·m on joints 5–7), **impedance\_residual** ’s continuous-envelope stress case.
- **derated\_joint4** — stock FR3 limits with joint 4 derated 87 → 31.5 N·m, **phri2** ’s own torque-activation stress case.

### 4.3 Diagnostics

Rows 1–4 follow the synthesis note’s §13.2 table. Row 1b is added.

| #  | Quantity                                                                                                    | Checks                           | Pass            |
|----|-------------------------------------------------------------------------------------------------------------|----------------------------------|-----------------|
| 1  | $\max_t \ \tau_{\text{base}} + J_v^\top F_{\text{nom}}\ _\infty / \bar{\tau}$<br>on the <b>phri2</b> replay | anchor feasible                  | $< 1$           |
| 1b | the same ratio, peaked over<br>the closed-loop $\alpha \rightarrow 0$<br>fallback run                       | anchor feasible <i>there too</i> | $< 1$           |
| 2  | $\lambda_{\min}(K_0), \lambda_{\min}(D_0)$                                                                  | passive                          | $> 0$           |
| 3  | settled $\ e_z\ $ at $\alpha \rightarrow 0$ under<br>the 20 N push                                          | fallback holds                   | $\leq 0.06$ m   |
| 4  | $\max_t \ J_v^\top F_{\text{nom}}\ _\infty / \bar{\tau}$                                                    | not budget-dominating            | $< \beta = 0.5$ |

**Why row 1b exists.** Rows 1 and 3 are measured at two different operating points. Row 1 replays the trajectory **phri2** ’s MPC actually produced, on which the EE stays near the nominal pose, so  $F_{\text{nom}} = -K_0 e$  is small whatever  $K_0$  is. Row 3 describes the *displaced* equilibrium the arm settles into once the residual is de-authorized, where by construction  $K_0 e = F_h$ . Merged Lemma 1 requires  $|a_\ell| \leq \bar{\tau}$  at **every** tick, and the fallback’s ticks are ticks. Row 1b closes that hole by peaking the same ratio over the fallback run itself — transient included, since Lemma 1’s precondition has no exemption for overshoot. Omitting it more than doubles the apparent feasible region (§5.2).

Rows 1 and 4 are replay diagnostics: cheap, and exact *given* that the merged controller’s total command tracks **phri2** ’s. That assumption is the main caveat on this result and is stated as such in §7.

Row 3 is measured on a closed-loop MuJoCo run of the actual  $\alpha \rightarrow 0$  law,  $\tau = \tau_{\text{base}} + J_v^\top F_{\text{nom}}$ , not from static algebra. The push hold is extended to 6 s and the displacement averaged over the final 1.5 s: at  $k = 380$  N/m against  $\Lambda_{zz} = 4.56$  kg the fallback spring has a 0.69 s natural period, so **phri2** ’s own 2 s hold and a sub-period averaging window read an oscillation rather than an equilibrium — and biased the estimate in the flattering direction, making low-damping cells look *better* than they are.

### 4.4 Grid

$k \in [10, 1200]$  N/m log-spaced,  $d \in [2, 60]$  N·s/m, densified through  $k \in [300, 520]$  N/m and  $d \in [8, 40]$  N·s/m:  $36 \times 25 = 900$  cells. The note proposes  $k \leq 400$  N/m and says to widen if the region sits at an edge. It does — row 3 alone forces  $k \gtrsim 350$  N/m — and the densification is not cosmetic: on a  $24 \times 16$  grid the region is  $\sim 40$  N/m tall and the scan reports it **empty** (§6.3).

## 5. Results

### 5.1 rho\_0.28 : infeasible at this pose

**Corrected by §9.2.** This section originally called the result structural. It is not: it is a property of **phri2** ’s nominal pose and push direction. At other poses **phri2** ’s own controller runs inside the  $\rho = 0.28$  envelope. Everything measured below is correct as measured; the word “structural” was an over-claim, and §9.2 gives the counter-example.

|                                                            | value          |
|------------------------------------------------------------|----------------|
| $\tau_{\text{base}}$ alone, as a fraction of joint 4's cap | <b>0.822</b>   |
| phri2 's own realized torque, as a fraction of this cap    | <b>1.300</b>   |
| cells passing row 1                                        | 197 / 900      |
| cells passing row 1b                                       | <b>0 / 900</b> |
| cells passing row 3                                        | 385 / 900      |
| <b>common region</b>                                       | <b>0 cells</b> |
| best worst-normalized use over the whole grid              | 1.238          |

Row 1b fails in **every cell**, with a grid-wide minimum of 1.144. This is not a tuning result *at this pose*. Holding a 20 N push at the fallback equilibrium requires  $K_0 e = F_h$ , so  $J_v^\top F_{\text{nom}} = -J_v^\top F_h$  *regardless of  $K_0$*  — the gains set where the arm sits, not how hard it must push back. On this pose that costs  $\sim 28.4$  N·m on joint 4 asymptotically (§6.2), against a 24.36 N·m cap. No  $(K_0, D_0)$  can change that.

The second row of the table is the more consequential one. phri2 's published FR3 trajectory peaks at 1.30× this envelope, so the envelope is not one the phri2 benchmark ever operated in. Carrying  $\rho = 0.28$  into the merge, as the synthesis note assumes, does not merely tighten the design problem — it changes the scenario to one in which the source result does not exist.

## 5.2 derated\_joint4 : feasible, thin

|                                                            | value                                              |
|------------------------------------------------------------|----------------------------------------------------|
| $\tau_{\text{base}}$ alone, as a fraction of joint 4's cap | 0.636                                              |
| phri2 's own realized torque, as a fraction of this cap    | 1.005                                              |
| cells passing row 1 / 1b / 3                               | 577 / 702 / 385                                    |
| <b>common region</b>                                       | <b>33 / 900 cells</b>                              |
| extent                                                     | $k \in [360, 400]$ N/m, $d \in [21.3, 48.4]$ N·s/m |
| region if row 1b omitted                                   | <b>70 cells</b>                                    |
| region if row 3 gated on peak rather than settled          | <b>6 cells</b>                                     |
| region after $\beta = 0.5$                                 | 33 cells (row 4 never binds)                       |

**Recommended operating point.** The note's literal criterion — minimize row 4 subject to rows 1–3 — selects  $k = 360$  N/m,  $d = 21.3$  N·s/m, which sits 0.4 % from violating row 1b. That is the wrong objective when the region is a sliver: row 4 is the only diagnostic the note does not hard-gate. The cell furthest from *every* hard constraint is the one to take:

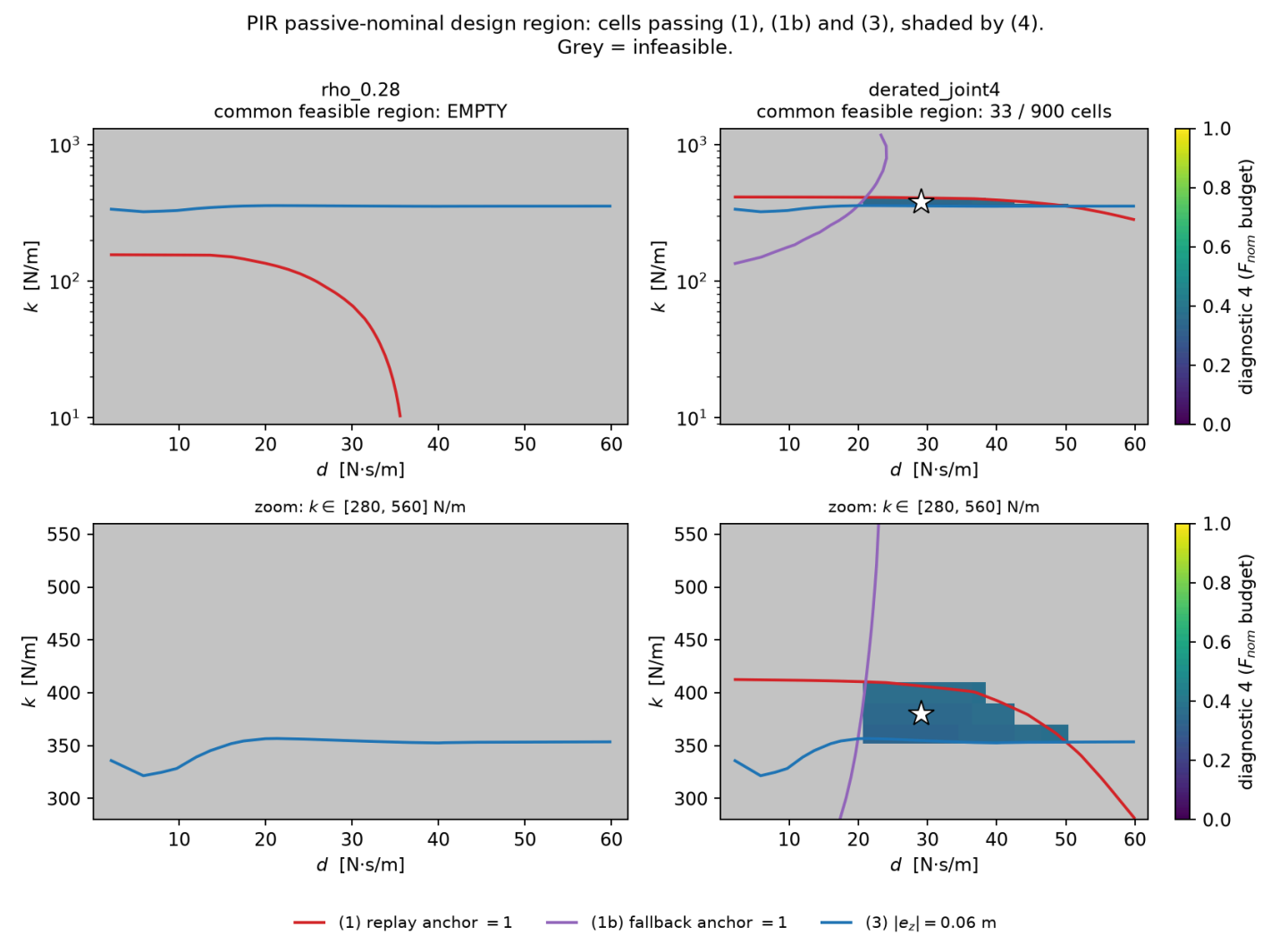
$$K_0 = 380 \text{ N/m}, \quad D_0 = 29.1 \text{ N} \cdot \text{s/m}$$

| diagnostic            | value          | limit         | use          |
|-----------------------|----------------|---------------|--------------|
| (1) replay anchor     | 0.977          | 1             | 98 %         |
| (1b) fallback anchor  | 0.976          | 1             | 98 %         |
| (3) settled $\ e_z\ $ | 55.8 mm        | 60 mm         | 93 %         |
| (3') peak $\ e_z\ $   | <b>62.2 mm</b> | 60 mm         | <b>104 %</b> |
| (4) budget fraction   | 0.348          | $\beta = 0.5$ | 70 %         |

This is a damping ratio of  $\zeta = 0.35$  at 1.45 Hz — a floor that is genuinely a floor, not a second behaviour layer.

**The three-way squeeze.** The overlay figure shows the region bounded below by row 3 (too soft and the fallback leaves the box), above by row 1 (too stiff and the anchor overruns joint 4 on the operating trajectory), and on the left by row 1b (too little damping and the fallback’s overshoot overruns joint 4). The note anticipated tensions (a)↔(c) and (c)↔(d). (d) never binds: row 4 ranges over 0.326–0.373 across the whole region, so  $\beta$  could be set anywhere above  $\sim 0.38$  without changing a single cell. The binding third constraint is the one the note did not list — damping against the fallback transient.

### 5.3 Figures



**Figure 1 — the design region.** Grey is infeasible. Left: the  $p = 0.28$  envelope, empty everywhere. Right: `phri2`’s derated-joint-4 envelope, 33 of 900 cells, with the recommended operating point starred. The zoom row is not decoration — the region is  $\sim 40$

N/m tall inside a 1190 N/m sweep, which is why a coarse grid reports it empty (§6.3). The three-way squeeze is visible in the lower-right panel: row (3) bounds it from below (too soft and the fallback leaves the box), row (1) from above (too stiff and the anchor overruns joint 4), row (1b) from the left (too little damping and the fallback’s overshoot overruns joint 4). The note anticipated the first two tensions; the third is the one it did not list.

| file                                                              | content                                                                                 |
|-------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| <code>pir_knot_scan_diag{1,1b,3,4}_{envelope}.png</code>          | the four diagnostic maps, per envelope                                                  |
| <code>pir_knot_scan_overlay.png</code>                            | composite, full range and zoomed, with the recommended cell marked                      |
| <code>pir_knot_scan.json</code>                                   | grid, all diagnostic arrays, <code>common_region_nonempty</code> , both recommendations |
| <code>pir_closed_loop_{scenario}_{envelope}.png/json</code>       | §7: the merged controller in closed loop                                                |
| <code>pir_e0_sweep.png/json</code>                                | §7.4–7.5: authorization vs tracking, both tightenings active                            |
| <code>pir_rootcause.png/json</code>                               | §8.1: every symptom swept against $K_0$                                                 |
| <code>pir_fixes.png/json</code>                                   | §8.3: candidate fixes scored on all four findings                                       |
| <code>pir_pose_study.png/json</code>                              | §9.1–9.2: the pose screen and its two traps                                             |
| <code>pir_knot_scan_pose_*</code>                                 | §9.3: the full gate re-run at the better pose                                           |
| <code>pir_closed_loop_pose_*</code>                               | §9.4: Task 2 re-run at the recommended pose                                             |
| <code>pir_e0_sweep_pose_*</code>                                  | §9.4: Task 3 re-run at the recommended pose                                             |
| <code>pir_axis_tension.png/json</code>                            | §10: which axis fires where, aggregated over every run                                  |
| <code>pir_rootcause_pose_*</code> , <code>pir_fixes_pose_*</code> | §8.4: §8 re-run at the recommended pose                                                 |
| <code>pir_joint_scan.png/json</code>                              | §10.5: the joint pose $\times K_0$ scan and its box-allowance sensitivity               |
| <code>pir_robustness.png/json</code>                              | §10.6: decision 3’s two formulations across 10 seeds $\times$ 2 profiles                |
| <code>pir_lemma_check.json</code>                                 | §8.5: Merged Lemma 1’ checked clause by clause                                          |

## 6. Verification

A NO-GO on one envelope and a 33-cell GO on the other are both strong enough claims to be attacked separately ( `pir_verify.py` , `results/pir_verify.json` ).

### 6.1 Check A — equilibrium without integration

Rows 1b and 3 are read off a settled simulation, and “settled” is a judgement call for a lightly damped spring. Check A instead solves  $\ddot{q}(q) = 0$  directly on MuJoCo’s own forward dynamics — no time integration at all — and compares. Across four cells the two methods agree to within **6.9 %** worst case (1.2 % at the well-damped end), all root-finds converged, all simulations flagged settled. The residual gap is the null-space posture spring, which makes the settled configuration mildly path-dependent; it is smaller than the 7 % margin the recommended cell carries on row 3, but not by much, which is part of why §7 calls that margin thin rather than adequate.

### 6.2 Check B — the fallback anchor does not depend on the gains

Sweeping  $k$  over  $[100, 1200]$  N/m at the root-found equilibrium:

| $k$ [N/m] | $\ e_z\ $ [m] | $k \ e_z\ $ [N] | joint-4 anchor [N·m] |
|-----------|---------------|-----------------|----------------------|
| 100       | 0.198         | 19.83           | 25.29                |
| 200       | 0.101         | 20.12           | 27.87                |
| 400       | 0.050         | 20.06           | 28.34                |
| 800       | 0.025         | 19.98           | 28.40                |
| 1200      | 0.017         | 19.95           | 28.39                |

$k \|e_z\|$  recovers the 20 N push to within 1 % across a  $12\times$  range of  $k$ , confirming  $K_0 e = F_h$  and therefore that the equilibrium anchor is a property of the push and the pose, not of the gains. It converges to  $\sim 28.4$  N·m — comfortably above the 24.36 N·m  $\rho = 0.28$  cap and comfortably below the 31.5 N·m derated-joint-4 cap. That single number is why the two envelopes give opposite answers.

As a side result, the  $k = 200$  row independently reproduces `phri2`’s own reported figure that a pure  $K_d = 200$  impedance gives 0.10 m of static displacement against the 0.06 m bound.

### 6.3 Check C — grid refinement

Re-scanning  $k \in [300, 500] \times d \in [8, 40]$  on a *linear* grid whose points deliberately do not coincide with the main scan’s:

| envelope                    | region         | best worst-normalized use |
|-----------------------------|----------------|---------------------------|
| <code>rho_0.28</code>       | empty (0 / 99) | 1.244                     |
| <code>derated_joint4</code> | <b>14 / 99</b> | 0.977                     |

Two things follow. The `derated_joint4` verdict and its best cell are reproduced independently (0.977 against the main scan’s 0.977). And the refinement is *necessary*: the original  $24 \times 16$  grid reported `derated_joint4` empty. The region is  $\sim 40$  N/m tall inside a sweep spanning 1190 N/m, and a grid that samples it at two points misses it. Any re-run of this gate at a different pose, push direction or envelope must re-densify before trusting an empty answer.

### 6.4 Regression suite

`test_pir_knot_scan.py`, 12 tests, all passing. They pin the constants to their sources, check the anchor arithmetic against the per-tick matrix product, assert the zero-gain limit reduces the anchor to  $\tau_{\text{base}}$ , verify the  $K_0 e = F_h$  relation on the plant, check the averaging window exceeds the fallback spring’s natural period, and assert the published grid stays dense enough through the feasible band.

## 7. The merged controller, implemented

Task 1’s gate was a *feasibility* argument; §7 of its first draft owed a closed-loop measurement. `pir_controller.py` implements the merged rule of §3 and `run_pir_closed_loop.py` runs it on the FR3.

### 7.1 What the merge actually required

Three things changed relative to grafting the tank onto `phri2`, and all three live in the QP:

- The nominal is closed-loop inside the horizon.**  $F_{\text{nom}}$  is state feedback, so the prediction runs on  $A - BG_0$  with  $G_0 = [K_0 \ D_0]$ . Freezing  $F_{\text{nom}}$  across the horizon would let the QP plan against a nominal its own plan invalidates.
- The torque rows became state-dependent.** In `phri2`,  $\tau_k = \tau_{\text{base}} + J_v^\top F_{\text{cmd},k}$  is affine in the decision variable alone. Here  $\tau_k = \tau_{\text{base}} + J_v^\top (-G_0 x_k + F_{r,k})$ , and  $x_k$  is itself affine in every earlier residual, so each torque row carries the accumulated nominal reaction.

3. **The behaviour input and the disturbance model had to be split.** `phri2` has one force signal driving both  $a_{id} = Cx + GF_h$  and the plant prediction, which is sound only when every external force is interaction force. Running under a disturbance, they are different signals. `control()` now takes `force_forecast` and `behaviour_forecast` separately, defaulting to equal — which recovers `phri2` exactly, and is asserted to. **This does not solve force misclassification**; deciding which measured force goes in which channel *is* that unsolved problem. It moves the assumption into a signature instead of hiding it in an equality.

One incidental gain: `phri2` 's QP has no principled infeasibility fallback and drops to its clipped reactive law. Here  $F_r = 0$  is exactly the  $\alpha \rightarrow 0$  passive nominal the Task 1 gate certified, so an infeasible solve degrades to a *certified* behaviour rather than an improvised one. (No solve was infeasible in any run reported here: 0 / 300 per run.)

## 7.2 Rows (1) and (4), re-measured in closed loop

This is the debt the first draft recorded. At  $K_0 = 380$  N/m,  $D_0 = 29.1$  N·s/m on `phri2` 's 20 N push:

|                      | replay estimate (Task 1) | merged closed loop | gap    |
|----------------------|--------------------------|--------------------|--------|
| row (1) anchor ratio | 0.9771                   | <b>0.9762</b>      | 0.09 % |
| row (4) budget ratio | 0.3478                   | <b>0.3474</b>      | 0.11 % |

The replay approximation holds. The gate's 2 % margin on row (1) survives, and the GO verdict does not need revisiting on this scenario. A regression test pins the two together at 1 % so the gate cannot silently drift from the controller it gates.

## 7.3 Merged Lemma 1 and the four-term residual

Across every run in §7.4–7.5 (36 sweep points plus 8 closed-loop runs), with the precondition holding, the conclusion held:  $|\tau_\ell| \leq \bar{\tau}$  at every one of 6000 ticks, with  $\alpha_\tau$  saturating the envelope exactly ( $\max |\tau|/\bar{\tau} = 1.0000$ ) rather than overshooting it.

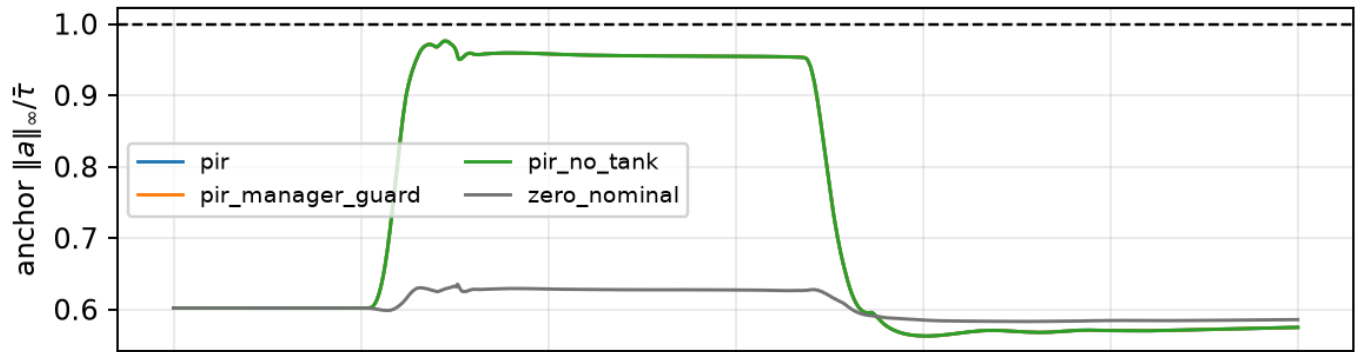
The four-term residual  $r = r_{\text{reg}} + r_{\text{con}} + r_{\text{mod}} + r_{\text{auth}}$  closes **to machine precision** ( $4.4 \times 10^{-16}$ ) on every QP tick, with  $r_{\text{con}}$  split into its plan-level and execution-level halves:

$$r_{\text{con}} = \underbrace{a_{\text{con}} - a_{\text{unc}}}_{\text{QP, 50 Hz}} + \underbrace{\Lambda^{-1}(\alpha_\tau - 1)F_{r,k}}_{\text{servo, 1 kHz}}, \quad r_{\text{auth}} = \Lambda^{-1}(\alpha_E - 1)\bar{F}_{r,\ell}.$$

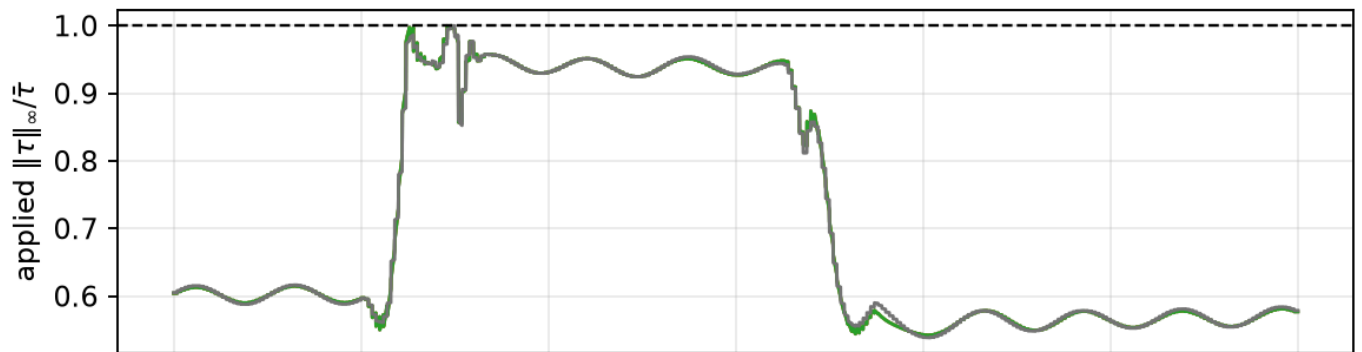
The closure is an identity, not a fit: the two servo terms telescope to  $\Lambda^{-1}(F_{r,\ell} - F_{r,k})$ , which is exactly the gap between what the QP proposed and what was applied.  $r_{\text{auth}}$  is identically zero whenever authorization never fires — it measures intervention, it is not an always-on correction.

# PIR merged controller, scenario 'merged', envelope 'derated\_joint4'

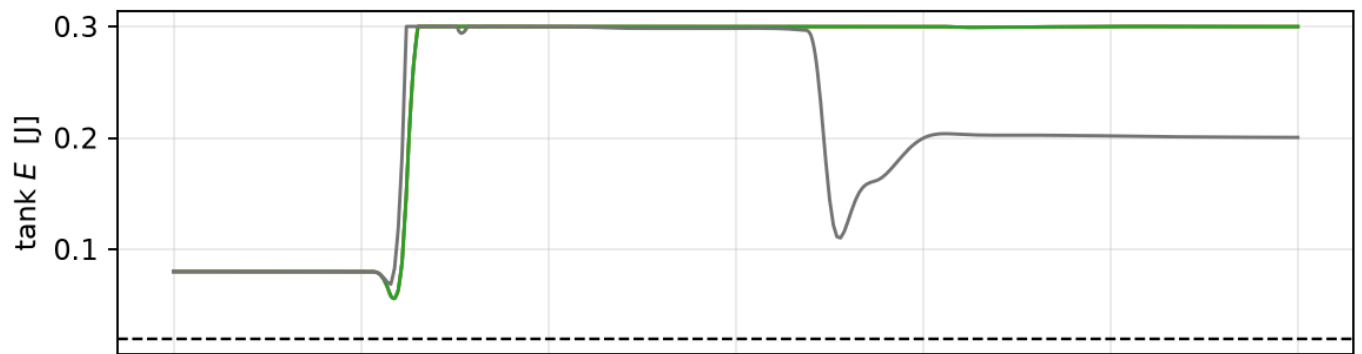
Merged Lemma 1's precondition (row 1), measured in closed loop



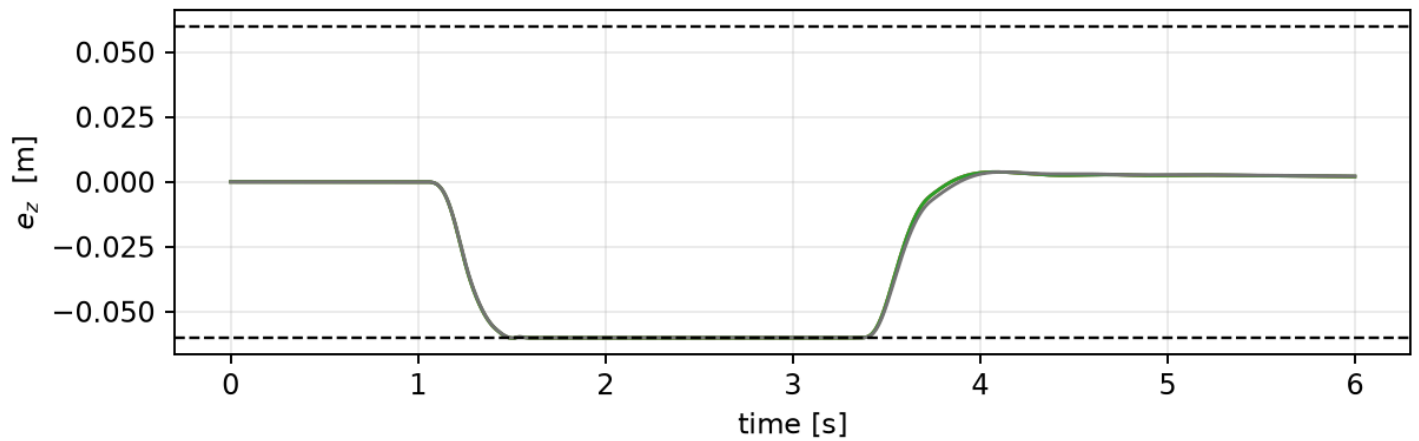
Merged Lemma 1's conclusion



energy tank vs its floor  $E_{\min} = 0.02$  J



workspace excursion on the push axis vs the 0.06 m bound



**Figure 2 — the merged controller in closed loop**, on the merged scenario at the recommended operating point. Top: Lemma 1’s precondition, the anchor ratio, sitting at 0.976 for the three PIR variants (they coincide here because authorization never fires at disturbance scale 1) against 0.635 for `zero_nominal` . That gap *is* \$7.6’s headroom collapse. Second panel: the conclusion — applied torque pinned at the envelope, never through it. Third: the tank, comfortably above its floor in this scenario, which is exactly why \$7.4 has to stress it deliberately. Bottom: the workspace excursion, slightly past the slack-relaxed 0.06 m box for every variant.

**7.4 The passivity axis: `impedance_residual` ’s result transfers**

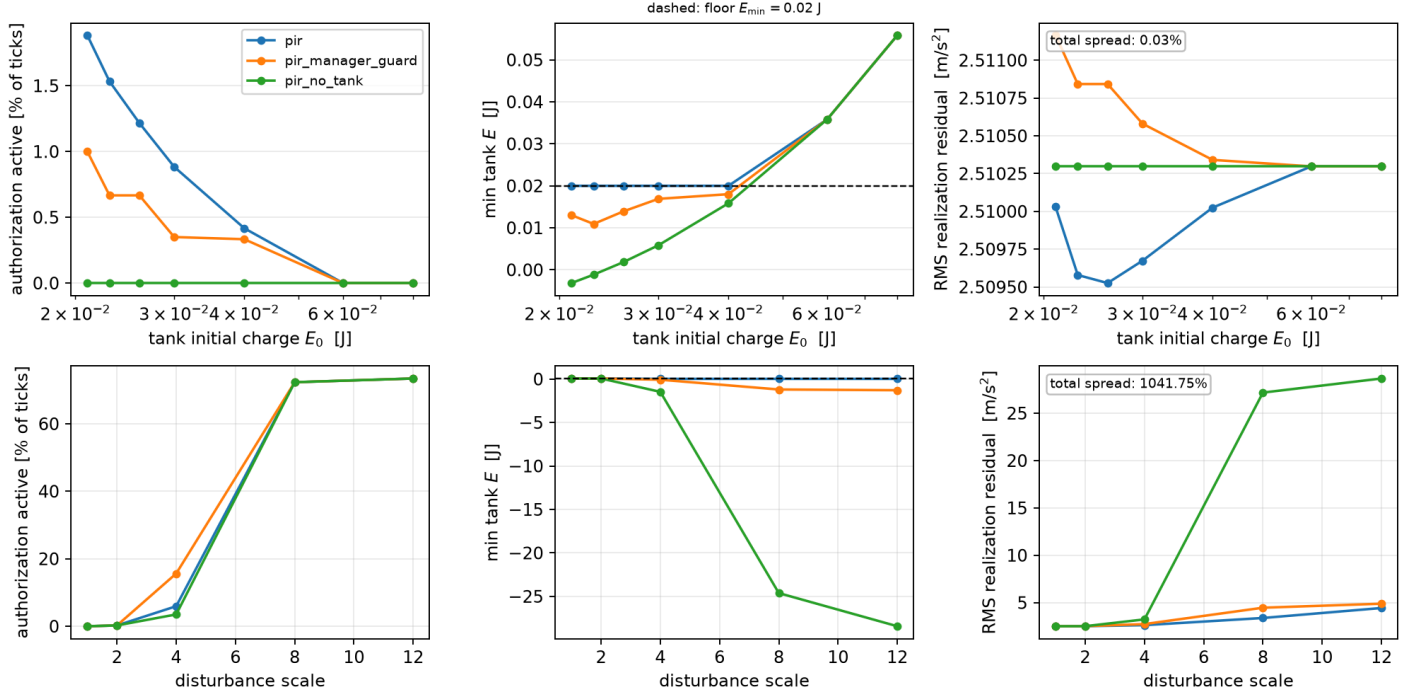
Neither source benchmark exercises both axes. `phri2` ’s push is monotone and largely dissipative: the nominal harvests  $v^T D_0 v$  faster than the residual drains the tank, and  $\alpha_E \equiv 1$  throughout. So the sweep runs on a **merged scenario** — that push plus `impedance_residual` ’s own rejectable disturbance (0.9/1.4/1.9 Hz sinusoids and a 12 N pulse placed deliberately between two 50 Hz manager ticks) — and sweeps  $E_0$  toward its floor and the disturbance amplitude upward. 36 points, three variants:

| variant                                      | tank floor held | breached ticks | worst tank            |
|----------------------------------------------|-----------------|----------------|-----------------------|
| <code>pir</code> (1 kHz re-authorization)    | 12 / 12         | 0              | 0.0200 J = $E_{\min}$ |
| <code>pir_manager_guard</code> (20 ms, held) | 4 / 12          | 6 189          | −1.285 J              |
| <code>pir_no_tank</code>                     | 4 / 12          | 13 050         | −28.43 J              |

*(Re-measured after decision 0 made  $\alpha_{\text{nom}}$  the default, which is why the breach counts are larger than the pre-decision-0 run: keeping the torque envelope legal at 16× disturbance leaves more energy for the ledger to have to account for. `pir` still holds the floor everywhere.)*

**`impedance_residual` ’s central result transfers to the merged port.** Fast re-authorization holds the floor in every stressed configuration — and holds it *with equality*,  $\min E = E_{\min}$  exactly, which is what the  $\alpha_E$  construction guarantees. The manager-rate guard tracks it closely and breaches anyway, in 8 of the 12 configurations where the tank is loaded at all. The 20 ms staleness is the whole difference.

Authorization goes from silent to active as either axis is loaded: 0 % → 1.9 % of ticks as  $E_0 \rightarrow E_{\min}$ , and 0 % → **73.4 %** of ticks as the disturbance scales 1 → 12. (That last figure counts any of the three scalings intervening, so past 4× it is dominated by  $\alpha_{\text{nom}}$ , decision 0’s fix, doing the work the anchor overrun used to do unchecked.)



**Figure 3 — authorization vs tracking, both tightenings active.** Top row sweeps the tank’s initial charge toward its floor; bottom row sweeps the disturbance amplitude. The middle column is the result: `pir` (blue) sits *exactly* on  $E_{\min}$ , which is what the  $\alpha_E$  construction guarantees; `pir_manager_guard` (orange) dips below it to  $-1.29$  J; `pir_no_tank` (green) reaches  $-28.4$  J. The right column carries its own spread annotation because matplotlib autoscales it — on the  $E_0$  axis the total variation is 0.03 %, i.e. flat (§7.5), and only on the disturbance axis does a real trade appear.

## 7.5 The authorization-vs-tracking trade looks flat here

**Corrected by §9.4.** This subsection originally concluded that the merged  $E_0$  trade is flatter than `impedance_residual`’s, against the synthesis note’s prediction that it would be steeper. That conclusion was a pose artifact. At the recommended pose the same sweep gives a **7.7 %** spread and the note’s prediction is confirmed. The measurement below is correct at this pose; the generalisation from it was not.

The synthesis note predicts the  $E_0$  curve gets steeper after the merge, because the two tightenings compound in series. Here it is flat: over the whole  $E_0$  sweep, RMS realization residual varies by **0.03 %** ( $2.5095 \rightarrow 2.5103$   $\text{m/s}^2$ ), against `impedance_residual`’s own  $21.49 \rightarrow 15.98$  mm swing on its unmerged nominal.

The reason is the §7.6 boundary. The passive nominal has already absorbed the authority the tank would otherwise take away: de-authorizing a residual that is 16 % of the command costs almost nothing, because the other 84 % is a PD law the tank has no say over. The passivity axis is *cheap* here precisely because it is *weak* — and §9.4 shows that when the residual is given room, the cost appears exactly as the note said it would.

On the disturbance axis, where the residual does matter, the trade appears even at this pose: RMS spans  $2.510 \rightarrow 4.434$   $\text{m/s}^2$  (77 %).

## 7.6 The nominal dominates the command, and eats the headroom

At the only  $(K_0, D_0)$  the gate certifies:

|                                 | pir                     | zero_nominal ( $K_0 = D_0 = 0$ ) |
|---------------------------------|-------------------------|----------------------------------|
| $\ F_{\text{nom}}\ $ RMS        | 14.05 N                 | 0 N                              |
| $\ F_r\ $ RMS                   | 2.11 N                  | 13.48 N                          |
| residual's share of the command | <b>15.6 %</b>           | 100 %                            |
| anchor headroom left for $F_r$  | <b>2.4 %</b> of the cap | 36.4 %                           |

The note’s §7 predicted this qualitatively — “ $K_0/D_0$  large  $\Rightarrow$  floor eats budget, a *new* double-tightening.” The measurement is worse than the phrasing suggests.  $K_0 = 380$  N/m is not a small passivity floor sitting under the behaviour: it is **stiffer than the desired impedance** ( $K_d = 200$  N/m), so the residual’s job is partly to make the arm *softer* than the floor. The merged controller at its certified operating point is a stiff PD with a 16 % predictive correction, and the residual’s torque headroom has collapsed **15-fold**, from 36.4 % of joint 4’s cap to 2.4 %.

### 7.7 Where the guarantee breaks

**Measured before decision 0, and fixed by it.** The table below is the controller without  $\alpha_{\text{nom}}$  — the variant now kept as `pir_no_nominal_auth`. It is what motivated §8.2’s fix. The shipped default holds  $\max|\tau|/\bar{\tau} = 1.0000$  at every scale tested, including the two that break here, so **the failure below is no longer the behaviour of the controller this document recommends**. It is retained because the precondition it exposes still fails at those scales —  $\alpha_{\text{nom}}$  keeps the envelope legal, it does not make the anchor feasible — and because §8.4 shows the fix is inert at the pose §9 recommends, so the failure mode is dormant rather than gone.

That 2.4 % headroom is what §1’s fourth finding cashes out. Sweeping the disturbance amplitude, without  $\alpha_{\text{nom}}$ :

| disturbance scale | max anchor ratio | ticks with infeasible anchor | max $\ \tau\ /\bar{\tau}$ |
|-------------------|------------------|------------------------------|---------------------------|
| 4                 | 0.9888           | 0 / 6000                     | 1.0000                    |
| 8                 | <b>1.0074</b>    | 39 / 6000                    | <b>1.0074</b>             |
| 12                | <b>1.0629</b>    | 558 / 6000                   | <b>1.0629</b>             |

Two things to read here. First, the applied-torque overrun **equals the anchor overrun exactly** at both failing scales. That is not a coincidence: when the precondition fails,  $\alpha_\tau$  has already gone to zero and the fast layer has *no remaining authority* — the overrun is entirely the nominal’s, and scaling the residual cannot touch it. Second, removing the tank as well changes nothing (1.0074 at scale 8 either way), confirming this is a property of the nominal and has nothing to do with the passivity axis.

So Merged Lemma 1 is sound and its precondition is doing real work: it is not a formality to be discharged once at design time. `phri2` guarantees the torque envelope unconditionally, because its QP owns the whole command. PIR guarantees it *conditional on the anchor being feasible*, and the Task 1 gate certified that condition against exactly one scenario. The condition does not survive an 8× disturbance. **This was the most important thing the implementation found**, and it is a cost of the split the synthesis note did not anticipate — though §8.2 fixes the consequence and §9 shows the pose largely removes the exposure, so §10 rather than this section is what ends up changing the paper’s claim.

## 8. Root cause, and what fixes it

Four findings is three too many if they share a cause. They do.

### 8.1 One constraint drives findings 1-3

Diagnostic 3 requires the  $\alpha \rightarrow 0$  fallback to hold the whole 20 N push inside 0.06 m, i.e.  $K_0 \geq |F_h|/0.06 = 333$  N/m. The behaviour the controller is supposed to render has  $K_d = 200$  N/m. **The gate therefore forces the passivity floor to be**

**stiffer than the behaviour it sits under**, and everything else follows. Sweeping  $K_0$  at fixed damping ratio  $\zeta = 0.35$

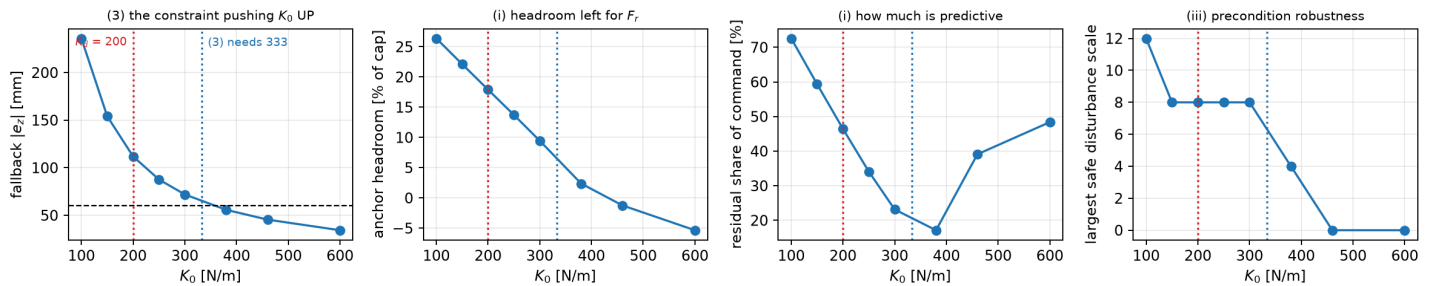
( pir\_rootcause.py ):

| $K_0$ [N/m]                   | fallback $\ e_z\ $ | (3) passes | anchor headroom | residual share | largest safe disturbance | $E_0$ trade   |
|-------------------------------|--------------------|------------|-----------------|----------------|--------------------------|---------------|
| 100                           | 236 mm             | no         | 26.4 %          | 72.6 %         | 12×                      | 0.0068        |
| 150                           | 154 mm             | no         | 22.1 %          | 59.4 %         | 8×                       | 0.0071        |
| <b>200 = <math>K_d</math></b> | 112 mm             | no         | 17.9 %          | 46.4 %         | 8×                       | 0.0125        |
| 250                           | 88 mm              | no         | 13.7 %          | 34.1 %         | 8×                       | 0.0358        |
| 300                           | 72 mm              | no         | 9.5 %           | 23.1 %         | 8×                       | 0.0057        |
| <b>380 = certified</b>        | 56 mm              | <b>yes</b> | <b>2.4 %</b>    | <b>17.1 %</b>  | <b>4×</b>                | <b>0.0020</b> |
| 460                           | 46 mm              | yes        | <b>−1.2 %</b>   | 39.1 %         | <b>0×</b>                | 0.0016        |
| 600                           | 34 mm              | yes        | −5.3 %          | 48.4 %         | 0×                       | 0.0010        |

Every symptom is monotone in  $K_0$ , every one of them gets worse as  $K_0$  rises, and diagnostic 3 is the only diagnostic that wants  $K_0$  large. At  $K_0 = 460$  the headroom is already negative — the precondition fails at the *nominal* disturbance. The certified band is thin from both sides.

**Is diagnostic 3's premise right?** It asks the passivity *floor* alone to satisfy a workspace bound that the desired *behaviour* does not satisfy either: phri2 's own  $K_d = 200$  gives 0.10 m of static displacement against the 0.06 m bound (reproduced here at 112 mm, §6.2), and phri2 meets the bound through its **predictive layer**, not through its impedance. Requiring more of the floor than of the behaviour is a defensible safety choice, but it is a *choice*, not a physical constraint — and \$8.3 prices it.

Root cause: every symptom is monotone in  $K_0$ , and only diagnostic (3) wants  $K_0$  large



**Figure 4 — the root cause.** Red dotted line: the desired impedance's own  $K_d = 200$  N/m. Blue dotted line: the 333 N/m diagnostic 3 demands. The certified operating point sits to the right of both, which is why the headroom and the precondition robustness are near their worst there. Only the leftmost panel improves as  $K_0$  rises.

## 8.2 Finding 4 is separate, and is fixed outright

Finding 4 — the fast layer has no authority over the nominal — is not a consequence of  $K_0$ . It is structural:  $\alpha_r$  scales  $F_r$  only. The fix is to give the servo an  $\alpha_{\text{nom}}$  as well, the largest scale keeping the anchor itself inside the box. Since  $\alpha_{\text{nom}} = 0$  recovers  $\tau_{\text{base}}$ , **the torque guarantee becomes unconditional whenever  $\tau_{\text{base}}$  alone fits** — a far weaker and checkable condition (0.636 of the cap here) than “the anchor fits at every future tick”.

It is not free, and the first attempt traded one guarantee for the other. A time-varying spring gain is an energy term, and only one direction is dangerous: *softening* releases stored energy (the storage function's  $\frac{1}{2}\dot{\alpha} e^\top K_0 e$  term goes negative, which helps), while *re-stiffening* is an injection — at fixed  $e$  the robot suddenly pushes back harder without the human having done the work to store

it. Charging re-stiffening to the tank and crediting softening nothing, unrestricted  $\alpha_{\text{nom}}$  buys the torque envelope and **loses the tank floor** (−1.06 J at 16×): a transient repeatedly re-buys stiffness the port has not earned.

Two rules fix it, and the difference between them turns out to matter more than the fix itself. *Forbidding* the rise —  $\alpha_{\text{nom}}$  **monotone non-increasing** — holds both guarantees out to 16× the source disturbance, against 4× for the certified baseline, and is what this document originally adopted. But forbidding is heavier than the problem needs, and §8.5’s (C5) shows why it is the wrong instrument: a monotone  $\alpha_{\text{nom}}$  is a **latch**, so one transient softens the nominal for the rest of the run.

*Metering* the rise does the same job without the latch. Re-stiffening may raise  $\alpha_{\text{nom}}$  by at most what the tank can afford at the current spring energy,

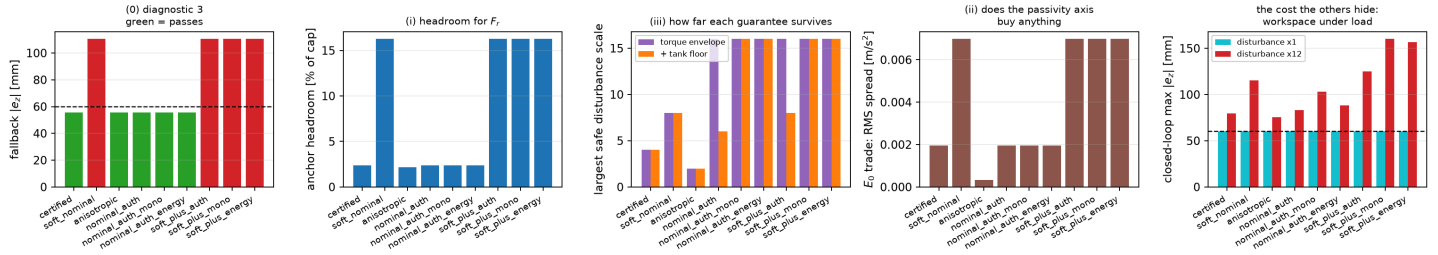
$$(\alpha_{\text{nom}} - \alpha_{\text{prev}}) \cdot \frac{1}{2} e^\top K_0 e \leq E - E_{\text{min}},$$

which is the same idiom as  $\alpha_E$  — authorize the injection, do not ban it — and is exactly the bound that makes the  $\max(0, \cdot)$  clamp inside  $\alpha_E$  non-binding, so the energy half still goes through (§8.5). Recovery is free when  $e^\top K_0 e \rightarrow 0$ , because nothing is stored and nothing is injected, and the rule degrades to the monotone one when the tank sits on its floor. `nominal_auth_energy` is the adopted default; `nominal_auth_mono` is kept as the ablation that shows what the metering buys.

8.3 Candidates, scored on all four findings

`pir_fixes.py` , all on the merged scenario, all against the same probes:

| candidate                | fallback | (3) | headroom | residual share | torque OK to | + tank floor to | $E_0$ trade | max $\ e_z\ $ at 12× |
|--------------------------|----------|-----|----------|----------------|--------------|-----------------|-------------|----------------------|
| certified $K_0=380$      | 56 mm    | ✓   | 2.4 %    | 17.1 %         | 4×           | 4×              | 0.0020      | 79 mm                |
| soft_nominal $K_0=K_d$   | 110 mm   | ✗   | 16.3 %   | 46.0 %         | 8×           | 8×              | 0.0070      | 115 mm               |
| anisotropic              | 56 mm    | ✓   | 2.2 %    | 20.0 %         | 2×           | 2×              | 0.0003      | 76 mm                |
| nominal_auth (free rise) | 56 mm    | ✓   | 2.4 %    | 17.1 %         | 16×          | 6×              | 0.0020      | 83 mm                |
| nominal_auth_mono        | 56 mm    | ✓   | 2.4 %    | 17.1 %         | 16×          | 16×             | 0.0020      | 103 mm               |
| nominal_auth_energy      | 56 mm    | ✓   | 2.4 %    | 17.1 %         | 16×          | 16×             | 0.0020      | 88 mm                |
| soft_plus_auth           | 110 mm   | ✗   | 16.3 %   | 46.0 %         | 16×          | 8×              | 0.0070      | 125 mm               |
| soft_plus_mono           | 110 mm   | ✗   | 16.3 %   | 46.0 %         | 16×          | 16×             | 0.0070      | 160 mm               |
| soft_plus_energy         | 110 mm   | ✗   | 16.3 %   | 46.0 %         | 16×          | 16×             | 0.0070      | 157 mm               |



**Figure 5 — candidate fixes.** The rightmost panel is the one that changes the recommendation: at nominal load every candidate holds the workspace box identically, and only under a 12× disturbance does the soft nominal’s real cost appear. Without that panel `soft_nominal` looks free.

Four things to take from this.

**`nominal_auth_energy` is a pure win and is what is adopted.** It quadruples the range over which *both* guarantees survive, costs nothing on any other metric, changes nothing at all while the anchor fits (asserted by test), and keeps diagnostic 3. There is no argument against it in these measurements.

**It strictly dominates the monotone rule it replaced.** Identical on every column above except the last, where the latch shows up as workspace: 88 mm against 103 mm at 12×, and 147 mm against 164 mm at 16×, over a baseline of 79 mm and 86 mm with no  $\alpha_{\text{nom}}$  at all. So metering the rise gives back **62 % of what the latch cost** at 12×, while holding the torque envelope at exactly 1.0000 and the tank at exactly  $E_{\text{min}}$  out to 16×. The free-rise variant is cheaper still (83 mm) and is not available: it loses the tank floor at 8× (−1.06 J at 16×).

**The `anisotropic` refinement — the plan’s own deferred “later refinement” — does not work here, and slightly hurts.** Headroom falls to 2.2 % and the safe disturbance range halves to 2×. The reason is visible in the trajectory: the displacement is 8:1 dominated by the push axis (max  $|e| = [7.5, 0.3, 60.3]$  mm), so softening the off-axis gains barely reduces  $\|J_v^T F_{\text{nom}}\|_{\infty}$  while removing the off-axis restoring force that was helping keep the arm near the pose. A negative result, but a clean one: the obvious lever is the wrong lever.

**Lowering  $K_0$  is not free, and the fallback bound is not the only thing it buys.** At nominal load every candidate holds the box identically (60.2 mm), which makes `soft_nominal` look costless. Under load it is not: at 12× disturbance the certified nominal holds 79 mm while `soft_plus_mono` reaches **160 mm**, and its RMS realization residual is worse too (5.60 vs 4.03 m/s<sup>2</sup>). So the real trade for findings 1–3 is not “56 mm vs 110 mm in a fallback that may never happen” — it is **workspace containment under disturbance**, which is a live property. That is a genuine engineering decision, and it is the human’s to make, not the scan’s.

## 8.4 Re-run at the recommended pose: the fixes solve a problem the pose removes

§8.1–8.3 were measured at `phri2`’s pose. Re-run at the recommended one (`pir_rootcause.py --pose`, `pir_fixes.py --pose`):

| $K_0$ | headroom ( <code>phri2</code> ) | headroom (rec.) | safe disturbance ( <code>phri2</code> ) | safe disturbance (rec.) |
|-------|---------------------------------|-----------------|-----------------------------------------|-------------------------|
| 100   | 26.4 %                          | 55.8 %          | 12×                                     | <b>12×</b>              |
| 250   | 13.7 %                          | 52.4 %          | 8×                                      | <b>12×</b>              |
| 380   | 2.4 %                           | 49.7 %          | 4×                                      | <b>12×</b>              |
| 460   | −1.2 %                          | 47.5 %          | <b>0×</b>                               | <b>12×</b>              |
| 600   | −5.3 %                          | 46.5 %          | <b>0×</b>                               | <b>12×</b>              |

The headroom never drops below 46 % at any  $K_0$ , and the feasibility axis never binds at any  $K_0$  — the safe disturbance scale is pinned at the top of the probe range throughout. The whole monotone-collapse story of §8.1 is a property of a tight torque

budget, and this pose does not have one.

The fix comparison collapses with it. All nine candidates of §8.3 now score **identically at the ceiling**: both guarantees hold to 16×, no tank deficit, and  $\alpha_{\text{nom}}$  **never fires for any of them**. Decision 0’s fix, which quadrupled the safe range at `phri2`’s pose, is entirely inert here.

That is not an argument against adopting it — insurance that never pays out in the tested scenario is still the right thing to carry, and §8.2’s failure mode is real wherever the budget *is* tight. But it does mean **§8 is a diagnosis of a configuration, not of the architecture**, and the paper should present it that way. Two of §8’s conclusions also need narrowing: the “passivity peaks at intermediate  $K_0$ ” shape inverts at this pose (§10.2), and `anisotropic` stops being harmful — 47.9 % headroom against `certified`’s 45.7 %, and the same 16× safe range, so at this pose it is merely useless rather than costly.

## 8.5 Merged Lemma 1’: what the fix actually guarantees, and what it costs

§10.6 exposed a gap between the lemma as stated and the controller as shipped. Across 20 resampled runs at §10.5’s cell, Lemma 1’s **precondition** held in 10 and its **conclusion** held in 20. Both cannot be right for a conditional guarantee: one whose condition fails half the time should fail half the time. What closes the gap is  $\alpha_{\text{nom}}$ , which is not in the lemma at all.

`pir_lemma_check.py` states the repaired lemma and checks every clause on 4000 adversarially sampled ticks ( $|e|$  to 0.35 m,  $|v|$  to 1.2 m/s,  $|F_r|$  to 600 N — far outside anything the QP emits), plus a scripted transient-and-return episode for the one clause adversarial sampling cannot reach.

### Merged Lemma 1’ (nominal authorization, energy-authorized re-stiffening).

*Hypotheses. (P1)  $|\tau_{\text{base},\ell}| \leq \bar{\tau}$  at every tick. (P2)  $E_0 \geq E_{\min}$ . (P3)  $\alpha_{\text{nom}}$  falls freely and rises only within the tank’s budget,  $(\alpha_{\text{nom},\ell} - \alpha_{\text{nom},\ell-1}) \cdot \frac{1}{2}e_\ell^\top K_0 e_\ell \leq E_{\ell-1} - E_{\min}$ . The monotone rule is the special case in which the left side is never positive, so everything below holds for it too.*

*Conclusions. (C1)  $|\tau_\ell| \leq \bar{\tau}$  for all  $\ell$ . (C2)  $E_\ell \geq E_{\min}$  for all  $\ell$ . (C3) until the first tick at which Lemma 1’s original precondition fails,  $\alpha_{\text{nom}} = 1$  and the controller is bit-identical to the one without it.*

**(P1) is the whole precondition, and it is much weaker than Lemma 1’s.** Lemma 1 required  $|\tau_{\text{base}} + J_v^\top F_{\text{nom}}| \leq \bar{\tau}$ , which depends on  $K_0$ ,  $D_0$ ,  $e$  and  $v$  — every design choice and the entire trajectory. (P1) depends on none of them: it is a property of the pose, and it uses 0.636 of joint 4’s cap at `phri2`’s pose and 0.433 at the recommended one. That is the real content of decision 0.

| clause                            | result over 4000 adversarial ticks                              |
|-----------------------------------|-----------------------------------------------------------------|
| (C1) torque envelope              | <b>0 violations</b> , worst $\ \tau\ /\bar{\tau} = 1.000000$    |
| (C2) tank floor                   | <b>0 violations</b> , worst $E = 0.020000 \text{ J} = E_{\min}$ |
| (C3) inertness from a fresh state | <b>0 violations</b> over 1326 applicable ticks                  |

**(P3) is load-bearing, not cosmetic.** The ledger debits  $\frac{1}{2}(\alpha_{\text{nom}} - \alpha_{\text{prev}})e^\top K_0 e$  when the nominal *re-stiffens*, and without (P3) nothing bounds that debit against the available energy. (P3) bounds it by  $E - E_{\min}$ , which is exactly what makes the  $\max(0, \cdot)$  clamp on the energy budget inside  $\alpha_E$  non-binding: with the clamp inactive the ledger lands at  $E_{\min}$  in the worst case rather than below it, whether the residual’s port power is positive (in which case  $\alpha_E$  throttles it to what is left) or not (in which case the ledger only gains). Drop (P3) — let  $\alpha_{\text{nom}}$  rise freely — and the same 4000 ticks give **3999 floor violations, worst  $E = -13\,395 \text{ J}$** . Both admissible rules give **0 violations, worst  $E = E_{\min}$  exactly**. So (P3) is not a tuning choice; it is what makes the energy half provable, and the choice *within* (P3) — forbid the rise or meter it — is what the next two clauses are about.

### (C5) — the latch, and how it was removed

Under the **monotone** rule  $\alpha_{\text{nom}}$  is monotone over the *whole run*, so once it fires it never recovers. The first version of this section stated (C3) without the “until the first tick”, and the checker falsified it on **1326 of 4000 ticks** — correctly. After the latch fires the controller is no longer inert even at ticks where Lemma 1’s precondition holds again: all 1326 differ.

The obvious repair is a per-contact reset, which §11 listed as owed. It is the wrong repair. An episode boundary is exactly the kind of thing the servo has no reliable way to detect, and a reset that fires on the wrong tick hands back stiffness the port has not

earned — the failure (P3) exists to prevent. *Metering* the rise is the same repair without the boundary, and it is the one adopted. Scripted episode — 200 ticks of transient, then a return with  $e$  decaying toward the reference:

| rule              | $\alpha_{\text{nom}}$ after the transient | tank             | recovery                               |
|-------------------|-------------------------------------------|------------------|----------------------------------------|
| monotone          | 0.851                                     | $E_{\text{min}}$ | never                                  |
| energy-authorized | 0.851                                     | $E_{\text{min}}$ | 25 ticks, at $\ e\  = 44.3 \text{ mm}$ |

The tank sits exactly on its floor when the transient ends, so *nothing* is affordable at that instant — the one-tick affordability threshold  $|e|^* = \sqrt{2(E - E_{\text{min}})/((1 - \alpha_{\text{nom}})K_0)}$  is 0 mm. Recovery does not come from a timer; it comes from the arm dissipating energy on the way back, and 25 ms of the return buys the whole of it. That is the property worth having: **the stiffness is re-earned, not re-granted.**

Two things this does not buy. First, in the adversarial sampler above  $\alpha_{\text{nom}}$  rises on 2151 of 4000 ticks but never reaches 1, and all 1326 inertness checks still differ. That is the rule working, not failing:  $|e|$  is held near 0.35 m throughout,  $\frac{1}{2}e^\top K_0 e$  is two orders of magnitude larger than the whole tank, and a robot held that far off its reference has not earned its stiffness back. Second, recovery is funded by dissipation, so a sequence of transients with little motion between them could keep  $\alpha_{\text{nom}}$  down indefinitely. Nothing here tests that horizon; §11 records it as what is still owed.

(C4) — the cost, and it re-opens decision 0

Where  $\alpha_{\text{nom}} < 1$  the rendered nominal is  $\alpha_{\text{nom}}K_0$ , so the storage is  $H_0 = \frac{1}{2}v^\top \Lambda v + \frac{1}{2}\alpha_{\text{nom}}e^\top K_0 e$  and the  $\alpha \rightarrow 0$  fallback displacement scales as  $1/\alpha_{\text{nom}}$ . Diagnostic 3’s guarantee — that the fallback holds the push inside the box — is **not** preserved:

| $\alpha_{\text{nom}}$ | effective $K_0$ | fallback $\ e_z\ $ | inside the 60 mm box |
|-----------------------|-----------------|--------------------|----------------------|
| 1.00                  | 380 N/m         | 55.8 mm            | yes                  |
| 0.75                  | 285 N/m         | 76.4 mm            | no                   |
| 0.50                  | 190 N/m         | 117.0 mm           | no                   |

That much is a property of the *authorization*. The **closed-loop** exchange rate, though, was mostly a property of the latch, and separating the two is the point of the middle column below (excursion as % of the 60 mm box,  $\max \|\tau\|/\bar{\tau}$  in brackets):

| disturbance | adopted (metered)     | monotone       | no $\alpha_{\text{nom}}$ |
|-------------|-----------------------|----------------|--------------------------|
| 1×          | 100 % [1.0000]        | 100 % [1.0000] | 100 % [1.0000]           |
| 8×          | <b>109 %</b> [1.0000] | 180 % [1.0000] | 109 % [ <b>1.0074</b> ]  |
| 12×         | <b>147 %</b> [1.0000] | 172 % [1.0000] | 132 % [ <b>1.0629</b> ]  |

Read the 8×

 row across. The monotone rule bought a 0.74 % torque overrun for **71 percentage points** of excursion — 108 mm instead of 65 mm against a 60 mm box — and that is the number the first version of this section reported as the price of decision 0. The metered rule buys the same envelope for **65.66 mm against the ablation’s 65.64 mm**: on this row the fix is free, and the 71 points were the ratchet, not the authorization. At 12× a real cost remains — 15 points of box for a 6.3 % overrun avoided — but it is a fifth of what was reported.

So the honest one-line reading of decision 0 is now:

$\alpha_{\text{nom}}$  converts a torque-envelope violation into a workspace-bound violation, and (P3) is what makes the energy half provable. Metering the re-stiffening rather than forbidding it keeps both and returns most of (C4)’s closed-loop cost.

The question this *used* to turn on — what  $\rho \tau_{\text{max}}$  actually means, since a 0.74 % overrun of a thermal budget is nothing while 43 mm of extra excursion is a lot — is answered in §12.0, and the answer is “neither thermal nor safety-certified”: both envelopes are labelled by their own sources as deliberately artificial stress budgets, with the FR3’s absolute limits kept as a separate backstop.

So the overrun the ablation incurs is over a chosen budget, at  $1.0074 \times 31.5 = 31.7 \text{ N}\cdot\text{m}$  — **36 % of the FR3's 87 N·m hardware limit**. It is not a safety event, and decision 0 never depended on it.

## 8.6 Propositions 2 and 3: recursive feasibility, scoped

§12.0 showed the missing recursive-feasibility theorem is load-bearing in two places at once — it decides whether the workspace box can be hard, *and* whether the controller can promise a solve at the next tick. It is worth finding out how much of it is actually missing. `pir_recursive_feasibility.py` states what can be proved about the merged QP and checks every clause against the rows the controller assembles, rather than against a re-derivation of them.

Reading the assembly, only three row groups are hard: the per-step torque rows, the residual magnitude box, and the residual rate rows. The workspace and speed rows carry free slack columns and cannot obstruct feasibility. That makes the zero-residual sequence a candidate certificate.

**Proposition 2 (feasibility certificate).** Let  $x^{\text{free}}$  be the  $\alpha \rightarrow 0$  rollout  $x_{k+1} = A_{cl}x_k + Bf_k + B_ad$  with  $A_{cl} = A - BG_0$ , and let  $\Delta_F$  be one step of the residual rate box (40 N here). If **(F1)**  $|F_{r,\text{prev}}|_\infty \leq \Delta_F$  and **(F2)**  $|\tau_{\text{base}} - J_v^\top G_0 x_k^{\text{free}}|_\infty \leq \bar{\tau}$  at every constrained step, then  $F_r \equiv 0$  with the slacks set to  $x^{\text{free}}$ 's box violations is a feasible point, so the QP is feasible.

**(F2) is not a new object — it is the Task 1 gate's diagnostic row 1b, verbatim.** Row 1b was added in §4.3 for what looked like a different reason (“require the precondition to hold on the fallback itself”) and cut the apparent region from 70 cells to 33. It turns out to be the QP's feasibility certificate, which explains the factor of two: half those cells could not certify their own fallback.

(F1) is the half worth dwelling on. When the held residual exceeds one rate step, zero is not reachable in one tick, and the certificate has to be the rate-limited **ramp** to zero (certificate B below). That is not bookkeeping: §7's documented fallback on an infeasible solve — “ $F_r = 0$  is exactly the  $\alpha \rightarrow 0$  passive nominal the gate certified” — sets the held residual to zero *at the servo*, which the QP's own rate row would not have permitted in one step. The ramp is the reachable version of that fallback.

**Proposition 3 (recursive feasibility, frozen model).** Let  $P$  solve  $A_{cl}^\top P A_{cl} - P = -I$  and let  $a_j$  be the  $j$ -th row of  $J_v^\top G_0$ . With  $x_{eq} = (I - A_{cl})^{-1}(Bf + B_ad)$ ,

$$X_f = \{x : (x - x_{eq})^\top P(x - x_{eq}) \leq c^*\}, \quad c^* = \min_j \left[ \frac{\bar{\tau}_j - |\tau_{\text{base},j} - a_j^\top x_{eq}|}{\sqrt{a_j^\top P^{-1} a_j}} \right]^2$$

is invariant under the  $\alpha \rightarrow 0$  rollout and contained in the torque polytope. If the frozen model is carried across two solves and the horizon terminal state lies in  $X_f$ , the previous solution shifted and padded with  $F_r = 0$  is feasible at the next tick.

Two things read straight off  $c^*$ . It is positive **iff (P1) holds strictly at the fallback equilibrium** — Merged Lemma 1's hypothesis doing a second job — and  $x_{eq}$  is diagnostic 3's fallback displacement, reached from the other direction. Three diagnostics introduced separately turn out to be three parts of one argument: diagnostic 3 says where the terminal set is centred, row 1b says the path there is torque-feasible, and  $X_f$  is the invariant set around it.

Checked over 1800 QP ticks — six closed loops, each tick's QP rebuilt, the candidate tested against its rows and the same QP solved alongside so the certificate can be paired with the solver's own verdict:

| pose               | dist. | (F1)   | (F2)   | either cert. | solver feasible | terminal in $X_f$ | $c_{\min}^*$ | hard box solves |
|--------------------|-------|--------|--------|--------------|-----------------|-------------------|--------------|-----------------|
| phri2              | 1×    | 100 %  | 98.7 % | 98.7 %       | 100 %           | 98.7 %            | 0.032        | 100 %           |
| phri2              | 4×    | 94.3 % | 95.7 % | 92.3 %       | 98.7 %          | 89.3 %            | <b>0</b>     | 92 %            |
| phri2              | 12×   | 75.3 % | 76.0 % | 71.7 %       | 95.7 %          | 74.0 %            | <b>0</b>     | 72 %            |
| <b>recommended</b> | 1×    | 100 %  | 100 %  | <b>100 %</b> | 100 %           | <b>100 %</b>      | 1.344        | 100 %           |
| <b>recommended</b> | 4×    | 100 %  | 100 %  | <b>100 %</b> | 100 %           | <b>100 %</b>      | 1.196        | 100 %           |
| <b>recommended</b> | 12×   | 100 %  | 100 %  | <b>100 %</b> | 100 %           | <b>100 %</b>      | 0.804        | 100 %           |

**The certificate is sound: 0 unsound ticks out of 1800.** It never called a QP feasible that OSQP rejected, which is the one thing a sufficient condition may not do and the check that would have caught a candidate construction drifting from the assembly.

**It is conservative, and that is the honest reading.** At `phri2`’s pose under 12× it fails on 28.3 % of ticks while the solver fails on 4.3 %. A failed certificate means “this argument does not apply here”, not “infeasible”.

**At the recommended pose every column is 100 %, at every disturbance out to 12×.** The certificate holds at every tick, the terminal set is comfortably nonempty ( $c^* = 0.8\text{--}1.5$  against  $10^{-2}$  at `phri2`’s pose), the horizon terminal state is always already inside it, and no solve is infeasible. So **Proposition 3 applies as stated there, and the terminal constraint that would make it self-enforcing could be added for free** — it never binds. At `phri2`’s pose it would bind on 1 tick in 4 under load, so adding it there would change the controller rather than document it.

**And a hard box would work at the recommended pose.** Pinning the slacks to zero and re-solving (25 probed ticks per run — the re-solve is the expensive part), the QP stays feasible at every probed tick at every disturbance — while at `phri2`’s pose it drops to 72 % at 12×. §12.0 took “hard boxes break this QP” from the source’s own comment; reproduced here, it is **pose-specific too**, and it only bites under load, which is why a nominal-load benchmark would never have found it. That does not license making the box hard — the source’s recursive-feasibility objection is about worst-case guarantees, not about one trajectory staying lucky — but it does narrow the claim.

Two numbers confirm the construction rather than the controller:  $\rho(A_{cl}) = 0.967$  at `phri2`’s pose and 0.926 at the recommended one, so  $A_{cl}$  is Schur and the Lyapunov step is legitimate; and  $x_{eq}$  lands at 55.0 mm against diagnostic 3’s independently computed 55.8 mm.

**What this does not prove.** Proposition 3 is about the QP’s own frozen model.  $\tau_{\text{base}}$ ,  $J_v$  and  $\Lambda^{-1}$  are re-frozen at every solve from the true nonlinear state, so the next QP does not start where this one predicted — the gap §7.1’s four-term residual measures. Nothing here closes it. What changes is its size: the missing step is now “the re-frozen model lands in  $X_f$ ”, not “feasibility, somehow”. And as with everything else in §8, the part that looks like a theory problem is largely a property of one pose.

## 8.7 What this does and does not settle

Findings 1–3 are one problem with a known knob and a priced trade. Finding 4 is solved. What remains open is the same thing §7.7 pointed at: even with  $\alpha_{\text{nom}}$ , the guarantee is “torque legal provided  $\tau_{\text{base}}$  fits”, and nothing here defends *that*. On this FR3 pose  $\tau_{\text{base}}$  uses 0.636 of joint 4’s cap with no control authority applied at all, so the margin is real but finite, and it is a pose property — which is why re-running the gate at a pose that does not load joint 4 remains the highest-value untried experiment (§10).

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## 9. The pose study: most of this was the scenario

Every number above comes from `phri2`’s nominal pose under its  $-z$  push. §8.4 flagged that the residual condition —  $\tau_{\text{base}}$  alone fitting the envelope — is a *pose property*, and that re-running the gate elsewhere was the highest-value untried experiment. It was.

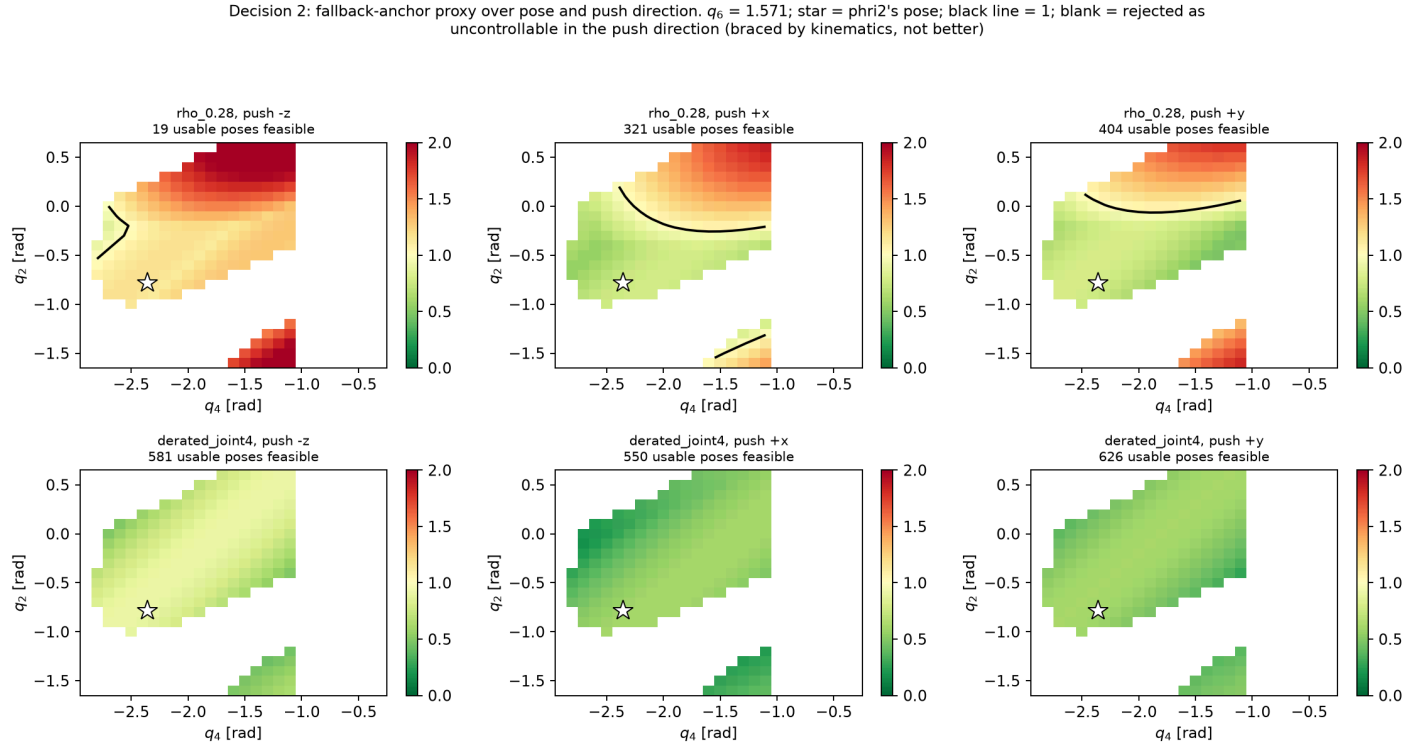
### 9.1 Screening poses, and two traps

`pir_pose_study.py` screens 1863 in-limit poses (varying  $q_2, q_4, q_6$ ; 1169 of them usable, i.e. end effector above the base plane and out in front) across three push directions, on two cheap quantities: the gravity floor  $\max |g(q)|/\bar{\tau}$ , and the fallback-anchor proxy  $g(q) - J_v^\top F_h$ , which is what the anchor tends to as  $K_0$  grows (§6.2).

**Trap 1: the proxy rewards poses that cannot render impedance.** Its first pick,  $(q_2, q_4, q_6) = (-0.20, -0.90, 2.60)$ , clears  $\rho = 0.28$  with an anchor ratio of 0.685 — and is near-singular. Its task-space inertia along the push axis is **47.2 kg** against the neutral pose’s 4.56 kg, and  $\sigma_{\min}(J_v)$  falls from 0.256 to 0.080. The arm is *braced*, not better: at 47 kg of apparent inertia the realization layer cannot render a 2 kg desired impedance in the push direction at all. A pose is therefore only a candidate if it also keeps the push direction controllable —  $\Lambda_{\text{axis}}$  within 2× of neutral and  $\sigma_{\min}$  at least 0.7× — and both thresholds are stored per pose so a different line can be drawn without re-running.

**Trap 2: the static proxy is optimistic, sometimes by a factor of three.** The gate uses  $\max_t \|\tau_{\text{base},t}\|_\infty/\bar{\tau}$  *along the trajectory*, where  $\tau_{\text{base}}$  also carries Coriolis terms and the null-space and orientation torques that become nonzero the moment  $q$

leaves  $q_{null}$ . One shortlisted pose scored **0.470 statically and 1.287 along the trajectory**, with the binding joint moving from 4 to 5. That is the other half of the lesson: relieving joint 4 can simply move the bottleneck to joints 5–7, whose  $\rho = 0.28$  caps are only 3.36 N·m. So the screen shortlists and a replay decides.



**Figure 6 — the pose screen.** Anchor proxy over  $(q_2, q_4)$  at  $q_6 = 1.571$ , per envelope and push direction. The star is phri2's own pose. Blank cells are rejected by the controllability guard — poses that resist the push through kinematic structure rather than control, which the unguarded screen preferred. Green is not yet a verdict: §9.1's second trap means these still have to survive an along-trajectory replay.

## 9.2 $\rho = 0.28$ is reachable — but not by enough

§5.1's "structurally infeasible" was too strong, and is corrected here. It is infeasible at phri2's pose and push direction. Elsewhere it is not: at  $(q_2, q_4, q_6) = (-0.10, -2.70, 1.571)$  under an  $+x$  push, phri2's own realized torque peaks at **0.952** of the  $\rho = 0.28$  cap — its published controller runs inside impedance\_residual's envelope there. The gain-independence argument of §6.2 stands; what was wrong was calling a scenario property a structural one.

It still does not help the merge. The best along-trajectory floor found anywhere in the shortlist is **0.870**, leaving  $\leq 13\%$  of the cap for  $J_v^\top F_{nom}$  — against the 36.4 % the neutral-pose derated envelope offers, which supported only a 33-cell sliver. A passive nominal does not fit in 13 %.

## 9.3 The derated envelope at a better pose: the actual result

Re-running the full gate at  $(q_2, q_4, q_6) = (-1.30, -1.30, 1.571)$ , keeping phri2's own  $-z$  push so only one thing changes:

|                               | phri2 's pose       | this pose           |
|-------------------------------|---------------------|---------------------|
| $\tau_{\text{base floor}}$    | 0.636               | <b>0.433</b>        |
| phri2 's own realized torque  | 1.005               | <b>0.508</b>        |
| cells passing row (1)         | 577 / 900           | <b>900 / 900</b>    |
| cells passing row (1b)        | 702 / 900           | <b>900 / 900</b>    |
| <b>common feasible region</b> | 33 cells            | <b>481 cells</b>    |
| most robust cell              | $K_0=380, D_0=29.1$ | $K_0=520, D_0=56.1$ |
| worst normalized use there    | 0.977               | <b>0.541</b>        |

And in closed loop, at each pose's own recommended operating point, sweeping the disturbance:

| disturbance | headroom<br>( phri2 pose) | headroom (this<br>pose) | residual share         | $\max \ \tau\ /\bar{\tau}$ | $\max \ e_z\ $            |
|-------------|---------------------------|-------------------------|------------------------|----------------------------|---------------------------|
| 1×          | 2.4 %                     | <b>45.7 %</b>           | 17.1 % → <b>54.0 %</b> | 1.0000 → <b>0.532</b>      | 60.2 → <b>51.5</b> mm     |
| 8×          | −16.8 %                   | <b>43.2 %</b>           | 72.4 % → 86.7 %        | 1.0000 → <b>0.533</b>      | 108.0 → <b>58.9</b><br>mm |
| 16×         | −55.5 %                   | <b>35.4 %</b>           | 85.0 % → 90.7 %        | 1.0000 → <b>0.619</b>      | 164.3 → <b>59.2</b><br>mm |

Three things follow.

**Findings 1 and 3 are largely gone.** The residual is 54 % of the command instead of 17 %, with 45.7 % of joint 4's cap behind it instead of 2.4 %.

**Finding 4 stops being stressed at all.** At 16×

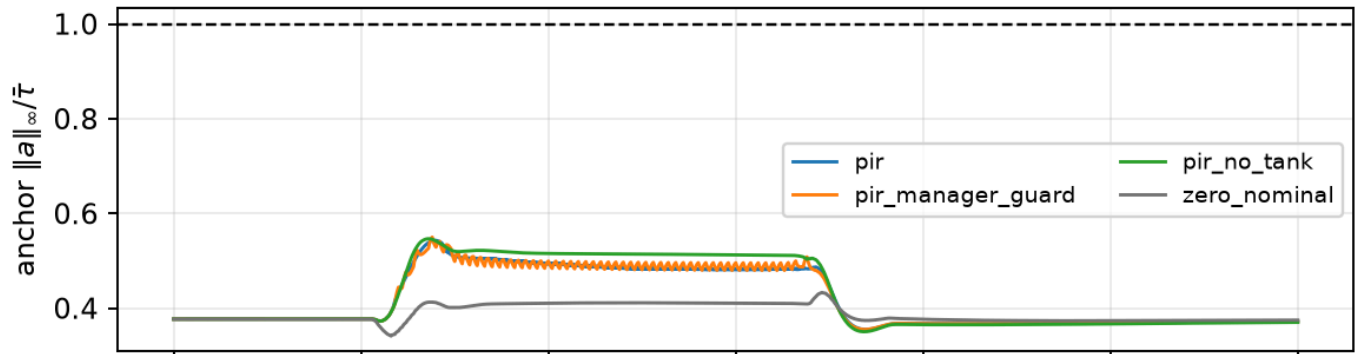
 disturbance the anchor still has 35 % headroom and the applied torque peaks at 0.62 of the cap — the controller is not saturating, so Lemma 1's precondition is never in question. Decision 0's  $\alpha_{\text{nom}}$  remains the right insurance (the negative headroom entries in the phri2 -pose column are exactly it absorbing an overrun that would otherwise have been a violation), but at this pose it is not called on.

**It is better on the workspace too, not traded against it.** 59 mm at 16×

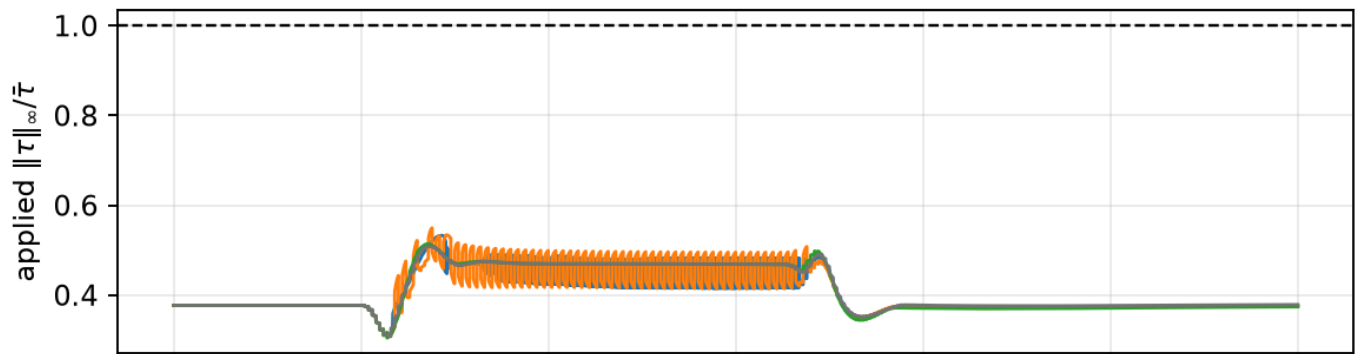
 disturbance against 164 mm — inside the 0.06 m box where the certified point is 2.7× outside it. This is the opposite of the  $K_0$  trade §8.3 priced, which bought headroom *with* containment. **Decision 1 may therefore be moot:** the pose change delivers what softening  $K_0$  was being considered for, and improves the thing softening  $K_0$  would have cost.

# PIR merged controller, scenario 'push', envelope 'derated\_joint4'

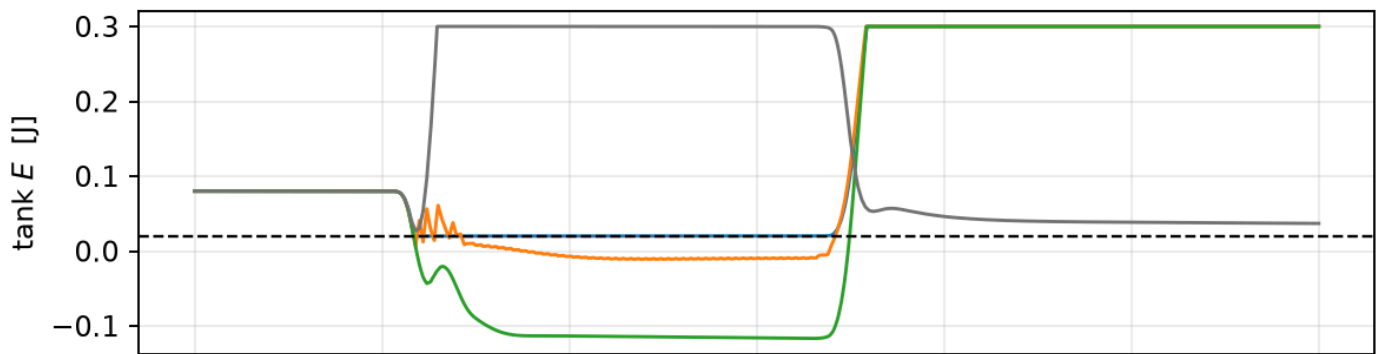
Merged Lemma 1's precondition (row 1), measured in closed loop



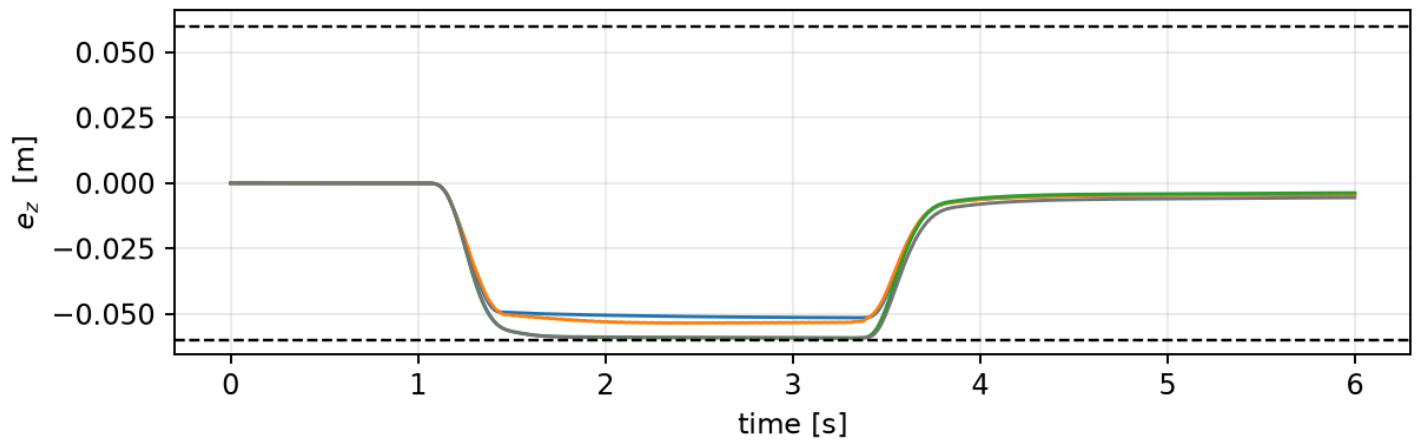
Merged Lemma 1's conclusion



energy tank vs its floor  $E_{\min} = 0.02$  J



workspace excursion on the push axis vs the 0.06 m bound



**Figure 7 — the merged controller at the recommended pose**, on `phri2` 's plain 20 N push. The third panel is the one to read: `pir` (blue) rides *exactly* on  $E_{\min}$  for the whole push — it is under the dashed line, which is the guarantee made visible — while `pir_manager_guard` (orange) sits below the floor for roughly two seconds and `pir_no_tank` (green) plunges to  $-0.117$  J. `zero_nominal` (grey) saturates at  $E_{\max}$  because with no nominal its residual *is* the impedance and is purely dissipative. Compare the top two panels with Figure 2: everything now runs at 0.4-0.55 of the envelope rather than pinned against it.

### 9.4 Tasks 2 and 3 re-run at the recommended pose

Everything in §7 and §8 was measured at `phri2` 's pose. The merged paper should report the pose it recommends, so both tasks were re-run at  $(q_2, q_4, q_6) = (-1.30, -1.30, 1.571)$  with that pose's own operating point,  $K_0 = 520$  N/m,  $D_0 = 56.1$  N·s/m.

**A bug surfaced first, and the four-term residual is what caught it.** The realization QP and the 1 kHz servo each build  $\tau_{\text{base}}$ , and the QP was centring the null-space posture spring on `Q_NEUTRAL` while the servo centred it on the actual pose — so the QP planned against a  $\tau_{\text{base}}$  and a  $d_{\text{known}}$  the servo never applied. At `Q_NEUTRAL` the two coincide and the bug is invisible; at the new pose the closure went from  $4 \times 10^{-16}$  to  $8 \times 10^{-5}$ . Fixed, closure back to  $1.3 \times 10^{-15}$ , and a regression test now pins it at both poses. This is the second time the closure identity has caught something no other diagnostic would have.

**Task 2 — the merged controller.** On `phri2` 's plain 20 N push:

|                               | <code>phri2</code> pose | recommended pose              |
|-------------------------------|-------------------------|-------------------------------|
| row (1), replay → closed loop | 0.9771 → 0.9762         | 0.5411 → <b>0.5432</b>        |
| anchor headroom               | 2.4 %                   | <b>45.7 %</b>                 |
| residual share of command     | 15.6 %                  | <b>53.5 %</b>                 |
| $\max \ \tau\ /\bar{\tau}$    | 1.0000                  | <b>0.5330</b>                 |
| authorization active          | 0.6 % of ticks          | <b>29.3 %</b>                 |
| $\min \alpha_E$               | 1.0000                  | <b>0.0006</b>                 |
| $r_{\text{auth}}$ RMS         | 0.0000                  | <b>1.387</b> m/s <sup>2</sup> |
| max excursion                 | 60.4 mm                 | <b>51.4 mm</b>                |

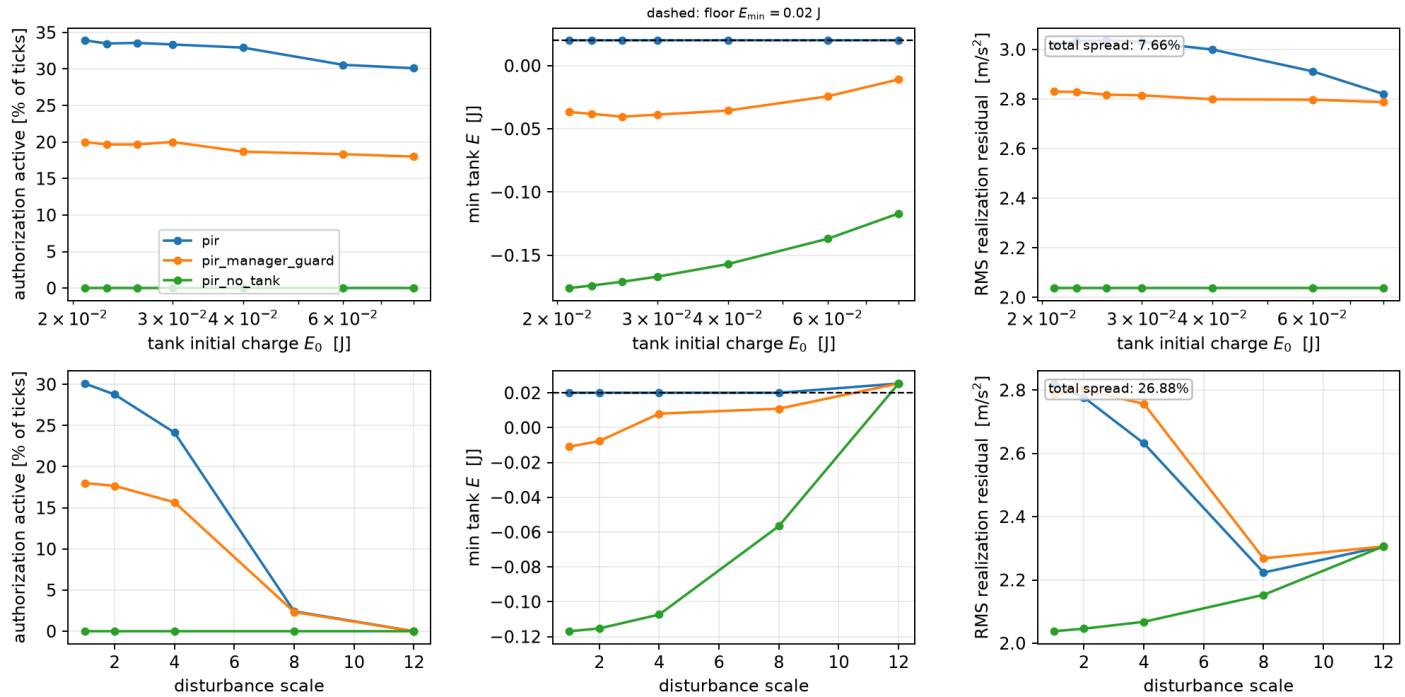
**The passivity axis is now exercised by `phri2` 's own benchmark.** §7.4 had to borrow `impedance_residual` 's oscillatory disturbance to make the tank do anything, and concluded that neither source benchmark stresses both axes. At the recommended pose that is no longer true: on the plain push, authorization fires on 29 % of ticks,  $\alpha_E$  reaches 0.0006, and `pir_manager_guard` breaches the floor for **2 039 ticks** while `pir_no_tank` reaches  $-0.117$  J. One scenario, both axes, no borrowed disturbance — which is what the merged paper needs.

**Task 3 — the authorization-vs-tracking trade, and a correction.** The  $E_0$  sweep at the recommended pose:

|                                           | <code>phri2</code> pose | recommended pose             |
|-------------------------------------------|-------------------------|------------------------------|
| $E_0$ trade, RMS spread                   | 0.03 %                  | <b>7.66 %</b>                |
| authorization active over the sweep       | 0.0-1.9 %               | <b>30.1-33.9 %</b>           |
| <code>pir</code> floor held               | 12 / 12                 | <b>12 / 12</b>               |
| <code>pir_manager_guard</code> floor held | 4 / 12                  | <b>1 / 12</b> (21 138 ticks) |
| <code>pir_no_tank</code> floor held       | 4 / 12                  | <b>1 / 12</b> (25 525 ticks) |
| tank's cost vs no tank at $E_0 = 0.08$    | +0.0 %                  | <b>+38.4 %</b> RMS           |

So §7.5's “flatter, not steeper” was a pose artifact and is corrected: the synthesis note’s prediction was right. Give the residual room and the passivity axis costs what the note said it would — 7.7 % across the  $E_0$  sweep, and 38 % against running with no tank at all. It is no longer cheap-because-weak; it is doing real work and charging real money for it. The fast-vs-manager-rate result also gets stronger, holding across the *entire*  $E_0$  range rather than only its bottom end.

PIR: authorization vs tracking with both tightenings active ( $K_0 = 520.0$  N/m,  $D_0 = 56.13$  N·s/m, envelope derated\_joint4)



**Figure 8 — authorization vs tracking at the recommended pose.** Against Figure 3, the middle column now shows `pir` holding  $E_{\min}$  across the whole sweep while both baselines breach everywhere, and the right column carries a 7.7 % spread instead of 0.03 % — a real trade rather than a flat line.

One counter-intuitive detail, stated because it is not obvious: authorization becomes *less* active as the disturbance grows (30.1 %  $\rightarrow$  0.0 % at 12 $\times$ ), the opposite of the `phr12` pose. The tank harvests  $\alpha_{\text{nom}} v^T D_0 v$ , which grows quadratically with the disturbance-driven velocity, while the residual’s debit  $F_r^T v$  grows linearly — so a large disturbance fills the tank faster than it drains it. The stressing case here is the *nominal* scenario, not the extreme one.

## 9.5 What this does not settle

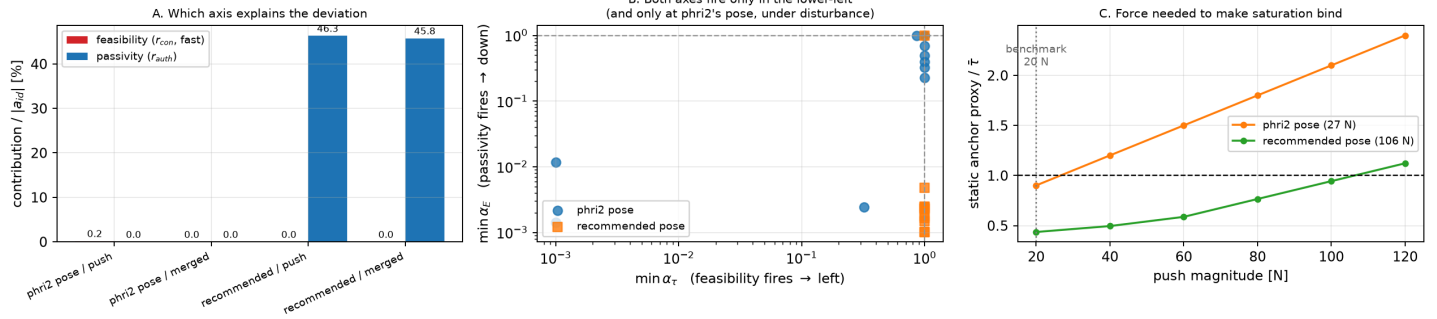
The pose was chosen by a screen tuned on one scenario, and only  $q_2, q_4, q_6$  were varied. Nothing here says it is optimal, or that it is a pose a real task would want the robot in — it is a better *test* pose, and whether the application permits it is a question this study cannot answer. The two guard thresholds are judgement calls. And the whole comparison is still one seed of one disturbance profile on one arm in simulation.

## 10. The tension between the two axes

PIR’s name rests on realizability having *two* axes — feasibility, reported by  $r_{\text{con}}$ , and passivity, reported by  $r_{\text{auth}}$  — and the synthesis note is explicit that without the second axis the whole thing is a rebrand of constrained-MPC impedance. So the question that decides the framing is not whether each axis works. It is whether there is an operating point where **both are load-bearing at once and the controller is well behaved**.

On the evidence of the runs collected up to §10.4, there is not; §10.5 then scans pose and  $K_0$  jointly and finds exactly one candidate, whose status depends on how the workspace box is read. `pir_axis_tension.py` aggregates every closed-loop run and sweep point in `results/` and classifies each by which authorization actually fired.

PIR's two axes are anti-correlated: each binds where the other does not



**Figure 9 — each axis binds where the other does not.**

## 10.1 Each axis is inert where the other is strong

Each axis's contribution is reported against  $|a_{id}|$ , the behaviour the controller is trying to render — the same denominator §10.5 uses, so the two are comparable. (Normalising by the *net* realization residual would not be a share of anything: the four terms sum to the net but oppose one another, so a single term can exceed it.)

| run                  | $\min \alpha_\tau$ | $\min \alpha_E$ | $r_{con}/  a_{id}  $ | $r_{auth}/  a_{id}  $ | headroom |
|----------------------|--------------------|-----------------|----------------------|-----------------------|----------|
| phri2 pose / push    | 0.8504             | <b>1.0000</b>   | 0.2 %                | <b>0.0 %</b>          | 2.4 %    |
| phri2 pose / merged  | <b>1.0000</b>      | <b>1.0000</b>   | 0.0 %                | <b>0.0 %</b>          | 2.4 %    |
| recommended / push   | <b>1.0000</b>      | 0.0006          | <b>0.0 %</b>         | 46.3 %                | 45.7 %   |
| recommended / merged | <b>1.0000</b>      | 0.0010          | <b>0.0 %</b>         | 45.8 %                | 45.7 %   |

At the recommended pose the passivity axis removes **46 %** of the intended behaviour and the feasibility axis contributes **nothing** —  $\alpha_\tau$  stays at 1.0 across all 12 sweep points there, so the fast torque projection never once intervenes. At `phri2`'s pose the situation inverts:  $\alpha_\tau$  fires and  $r_{auth}$  is *exactly* zero.

Across all 24 sweep points, **4 have both axes firing** — and all four are at `phri2`'s pose under 4–12× disturbance, which is precisely the regime §9.3 showed to be badly behaved (negative headroom, 103–108 mm excursion against a 60 mm box). The one well-conditioned dual-axis point,  $E_0 = 0.026$  at nominal disturbance, has  $\alpha_\tau = 0.9999$  — the feasibility axis is technically firing and doing nothing.

## 10.2 Why — and one place where I over-generalised again

Both authorizations act on the **same object**, the residual  $F_r$ , through the **same** torque budget. When that budget is tight they compete: a nominal stiff enough for saturation to bind squeezes the residual (17 % of the command at  $K_0 = 380$ , `phri2` pose), and a small residual does little port work, so the tank never drains.

The  $K_0$  sweep at `phri2`'s pose shows exactly that:

| $K_0$      | headroom | residual share | feasibility: safe disturbance | passivity: $E_0$ trade         |
|------------|----------|----------------|-------------------------------|--------------------------------|
| 100        | 26.4 %   | 72.6 %         | 12×                           | 0.0068                         |
| 200        | 17.9 %   | 46.4 %         | 8×                            | 0.0125                         |
| <b>250</b> | 13.7 %   | 34.1 %         | 8×                            | <b>0.0358</b> ← strongest here |
| 380        | 2.4 %    | 17.1 %         | 4×                            | 0.0020                         |
| 600        | −5.3 %   | 48.4 %         | <b>0×</b>                     | 0.0010                         |

I first wrote that up as a structural law — passivity peaks at intermediate  $K_0$ , feasibility binds monotonically — and the re-run at the recommended pose (§8.4) contradicts it, which is the same over-generalisation §11.1 catalogues, committed while writing the section that catalogues it. At the recommended pose the headroom never falls below 46 % at *any*  $K_0$ , the feasibility axis never binds at any  $K_0$ , and the  $E_0$  trade *rises* with  $K_0$  (0.021 up to  $K_0 = 380$ , then 0.32 at 460) instead of peaking in the middle.

What survives is narrower and still sufficient:

- **The competition is real only when the torque budget is tight.** With ample headroom a stiff nominal does *not* squeeze the residual — at the recommended pose  $K_0 = 460$  still leaves the residual 50 % of the command — so the two axes decouple. But they decouple into a regime where feasibility never binds at all, which is not a dual-axis regime either.
- **The axis-firing evidence is pose-robust and is what §10.1 rests on**, since it is measured directly rather than inferred:  $\alpha_r$  never once fires across 12 sweep points at the recommended pose, and  $r_{\text{auth}}$  is exactly zero across the headline runs at `phri2`’s.
- **Only a large disturbance loads both at once**, because it raises velocity (draining the tank) and the anchor (loading the envelope) together. That is why all four dual-axis points are high-disturbance ones.

So the conclusion of §10.1 stands — no well-behaved dual-axis operating point exists in the data — but the tidy mechanism I reached for does not, and §10.5’s joint scan is the way to find out which story is right.

### 10.3 The recommended pose optimises away PIR’s own feasibility half

Panel C quantifies it. Taking the static anchor proxy of §9.1 — which is *optimistic* about feasibility, so these are lower bounds:

| pose               | $\tau_{\text{base}}$ floor | push needed for the anchor to reach $\bar{\tau}$ |
|--------------------|----------------------------|--------------------------------------------------|
| <code>phri2</code> | 0.602                      | <b>27 N</b>                                      |
| recommended        | 0.377                      | <b>106 N</b>                                     |

The benchmark push is 20 N, which is already a firm two-handed shove. At the recommended pose you would need roughly **five times** that before saturation became the binding constraint — outside the range a human interaction produces.

So the pose that makes the passive-nominal split work is a pose where the saturation story PIR inherits from `phri2` does not arise. That is an uncomfortable sentence and it belongs in the paper, not in a footnote.

### 10.4 What this means for the framing

It does **not** mean the second axis is unearned. It means the claim has to be stated as what the evidence supports, which is a *stronger* claim than “both axes always bind”:

*The value of a two-axis certificate is not that both axes bind together. It is that **you cannot tell in advance which one will bind**. This study changes one FR3 joint angle — a change a practitioner would consider innocuous, and which improves every behavioural metric — and the binding axis flips from feasibility to passivity. A single-axis certificate is silent after that flip: at*

the recommended pose `phri2`'s  $r_{\text{con}}$  reports zero while half the realization deviation is energy authorization, and at `phri2`'s pose `impedance_residual`'s tank reports zero while the torque projection is the only thing intervening.

That is what the four-term residual buys, and it is demonstrable on the data already collected. The honest presentation is two case studies — a saturation case at `phri2`'s pose and a passivity case at the recommended one — with §10.2 as the explanation for why one experiment cannot be both.

## 10.5 The joint scan: one candidate, and it hangs on a judgement call

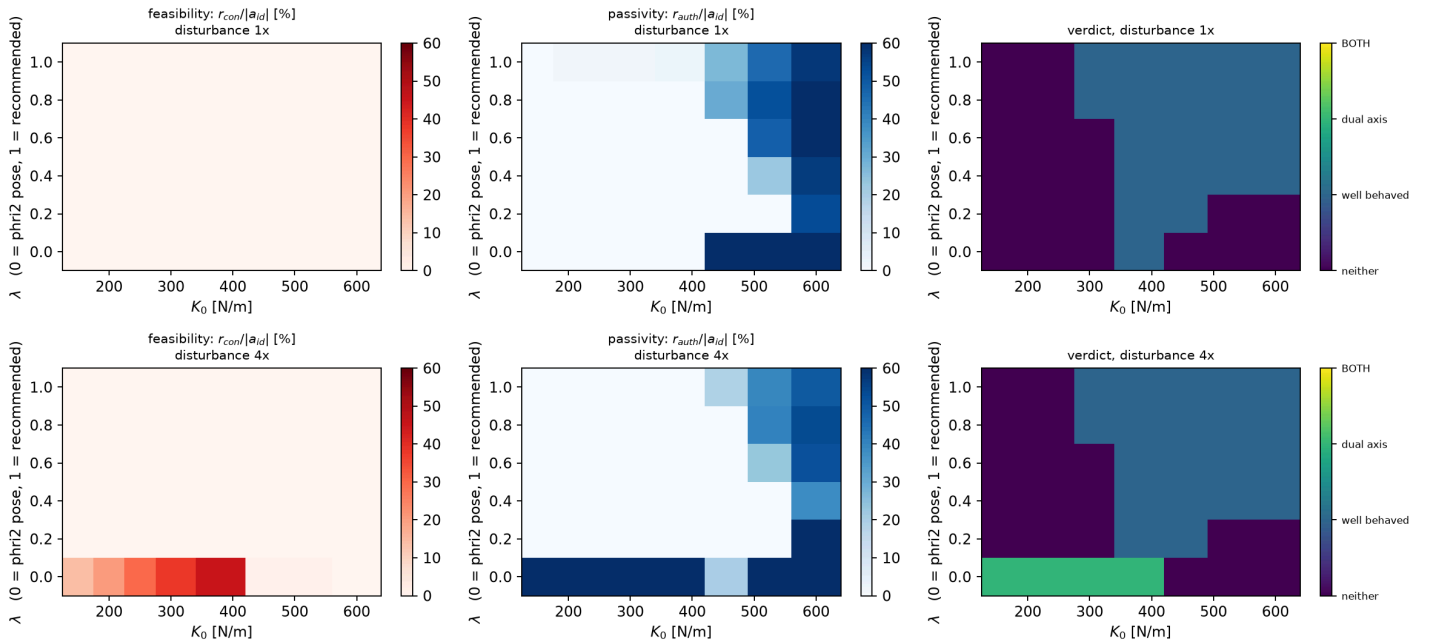
Pose and  $K_0$  had only ever been swept separately, so the two sweeps could have been cutting across a diagonal ridge neither resolved. `pir_joint_scan.py` scans them jointly: a pose family interpolating in joint space between `phri2`'s pose ( $\lambda = 0$ ) and the recommended one ( $\lambda = 1$ ), crossed with  $K_0 \in [150, 600]$  and disturbance  $\in \{1\times, 4\times\}$  — 96 cells, with  $D_0$  set from a fixed damping ratio against each pose's own task-space inertia so  $K_0$  is the only gain varying.

“Load-bearing” is deliberately stricter than “fired once”: an axis counts only if its residual term removes at least 2 % of the intended behaviour  $|a_{id}|$ . (Reported as a *ratio*, not a share — under a  $4\times$  disturbance the QP's residual fights something much larger than  $a_{id}$ , so the ratio legitimately exceeds 1. An earlier version of this script normalised by the *net* realization residual, which is not a share at all: the four terms sum to the net but oppose one another, and it produced “shares” of 468 %.)

**Result: 41 cells are well behaved, 5 are dual-axis, and the intersection is empty at the headline box allowance.** In every one of the 41 well-behaved cells  $r_{\text{con}}$  is *identically zero* — even in the three with under 5 % anchor headroom, where  $\alpha_\tau$  still never fires because the tight budget has squeezed the residual small enough to fit in what is left. That is §10.2's surviving mechanism, confirmed on a grid rather than inferred from two points.

All 5 dual-axis cells sit at  $\lambda = 0$  — `phri2`'s own pose — under  $4\times$  disturbance. And they fail “well behaved” for a reason worth stating precisely: **not because the certificate breaks**. Lemma 1's precondition holds, its conclusion holds, the tank floor holds, and  $\alpha_{\text{nom}}$  never fires in any of them. They fail on **workspace containment** — 69.5 to 134.7 mm against the 0.06 m box.

Joint pose x  $K_0$  scan: is there a well-behaved dual-axis operating point?



**Figure 10 — the joint pose x  $K_0$  scan.** Left column: the feasibility axis, which has colour only along the bottom edge ( $\lambda = 0$ , `phri2`'s pose). Middle: the passivity axis, concentrated at high  $K_0$  and high  $\lambda$ . The two never light up together anywhere the verdict panel (right) calls well behaved — no cell is marked “BOTH”. The one candidate of the next subsection is at  $\lambda = 0$ ,  $K_0 = 380$ , and it is classified here as dual-axis but not well behaved, because the 10 % box allowance this figure uses excludes it.

The candidate

That makes the verdict turn on how much overshoot of a **slack-relaxed** box counts as a violation, which is a judgement call the scan should not make silently:

| box allowance | dual-axis <i>and</i> well behaved                       |
|---------------|---------------------------------------------------------|
| 1.00× - 1.15× | <b>0</b>                                                |
| 1.20× - 1.30× | <b>1</b> — $\lambda = 0$ , $K_0 = 380$ , 4× disturbance |

At a 20 % allowance exactly one cell qualifies, and it is a striking one: `phri2` 's **own pose at the originally certified operating point**,  $(K_0, D_0) = (380, 29.07)$  — the point this whole study started from — under 4× the source disturbance:

|                                    |                                    |
|------------------------------------|------------------------------------|
| $r_{\text{con}}/\ a_{id}\ $        | <b>45.6 %</b>                      |
| $r_{\text{auth}}/\ a_{id}\ $       | <b>95.3 %</b>                      |
| $\min \alpha_\tau / \min \alpha_E$ | <b>0.320 / 0.0024</b>              |
| $\alpha_{\text{nom}}$              | 1.0 (decision 0's fix not needed)  |
| anchor headroom                    | +1.1 %                             |
| Lemma 1 precondition / conclusion  | hold / hold                        |
| tank floor                         | holds                              |
| fallback $\ e_z\ $ (diagnostic 3)  | 55.8 mm ✓                          |
| excursion                          | <b>69.5 mm</b> , 16 % over the box |

Both axes carrying 46 % and 95 % of the intended behaviour, both  $\alpha$ 's deep into intervention, and every certificate intact. If `phri2` 's box is read as what it is in its own QP — slack-relaxed, a preference under model mismatch rather than an actuator limit — this is the dual-axis demonstration the paper needs, and §10.1's "no such point" becomes "exactly one, and it is the obvious one".

**This promoted a question that was already on the owed list to a decisive one, and §12.0 answers it.** §12 had carried "resolve whether the 0.06 m bound is hard or slack-relaxed" since the Task 1 gate, where it decided a 33-cell versus 6-cell region; here it also decides whether this cell counts as an excursion or a violation. Both are asking the same thing — whether the box is a specification or a preference — and the source settles it: the box is slack-relaxed *by necessity*, because a hard box has no recursive-feasibility guarantee in this architecture and was empirically confirmed to make the QP infeasible.

So this cell is admissible, and it still does not carry the claim. §12.0 spells out why: the box reading removes one of the two objections to (b), and the other — 5 of 20 under resampling — is the one that decided it.

What it does not buy

One cell, at 4× the source disturbance, 16 % outside the nominal box, at one seed on one arm. It is an existence proof, not an operating recommendation — and note that it is *not* the pose §9 recommends. The honest reading is that PIR's two axes can be made simultaneously load-bearing, but only by running the arm harder than either source paper does, at the pose with the tight torque budget, and accepting an excursion the recommended pose avoids.

10.6 Both formulations, tested across seeds

§12's decision 3 offered two ways to state the claim, and both rested on a single disturbance seed — (b) on a single cell at a single seed. `pir_robustness.py` re-runs both across 10 seeds × 2 disturbance profiles (with and without `rejectable_force` 's between-manager-tick pulse, the component `impedance_residual` added specifically to defeat a slow guard). 60 runs.

Decision 3: robustness of formulation (a) the flip, and (b) the dual-axis candidate

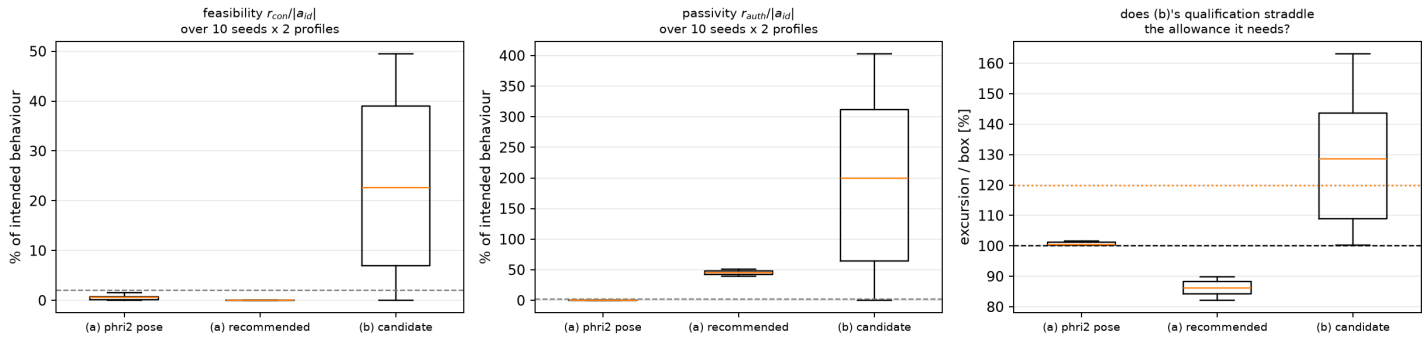


Figure 11 — decision 3 under resampling.

**(a) The flip survives, 20/20 — with one correction to how it is stated**

|                       | phri2 pose                       | recommended pose       |
|-----------------------|----------------------------------|------------------------|
| $r_{auth}/  a_{id}  $ | exactly 0 in 20/20               | 39.8 - 50.6 % in 20/20 |
| $r_{con}/  a_{id}  $  | 0.00 - 1.51 %                    | exactly 0 in 20/20     |
| classification        | neither axis load-bearing, 20/20 | passivity-only, 20/20  |

The assignment never swaps. But the strict load-bearing threshold exposes something §10.4's wording glosses: at  $1\times$  disturbance the feasibility axis **never reaches 2 % of the intended behaviour at either pose** — it fires ( $\alpha_\tau$  ranges 0.77-1.00 at phri2 's pose) but what it removes is under 1.5 %. So the robust statement is not “the load moves from one axis to the other”. It is:

*The passivity axis goes from **exactly zero** to **40-51 %** of the intended behaviour, and the feasibility axis from small-but-nonzero to **exactly zero**, on a joint-angle change that improves every behavioural metric. What flips is **which axis is capable of carrying load at all**.*

That is still the claim §10.4 wants and it is now 20/20 rather than 1/1. It is weaker than “the axes exchange the load”, and the paper should say the weaker thing.

**(b) The exhibit does not survive as stated**

|                                               | result                 |
|-----------------------------------------------|------------------------|
| dual-axis <b>and</b> every certificate intact | 5 / 20                 |
| Lemma 1 precondition holds                    | 10 / 20                |
| Lemma 1 conclusion holds                      | 20 / 20                |
| tank floor holds                              | 20 / 20                |
| excursion, all runs                           | 100 - 163 % of the box |
| excursion, runs that qualify                  | 111 - 120 %            |

§10.5's cell was seed 0 with the pulse — one of the lucky draws. Resampled, the candidate is dual-axis in 15 of 20 runs but keeps **all** its certificates in only 5, and the failure is always the same one: **Lemma 1's precondition**, which holds in exactly half the runs. (The conclusion and the tank floor never fail —  $\alpha_{nom}$  and the fast re-authorization do their jobs every time. That is a real result in itself, and it is decision 0's fix earning its place.)

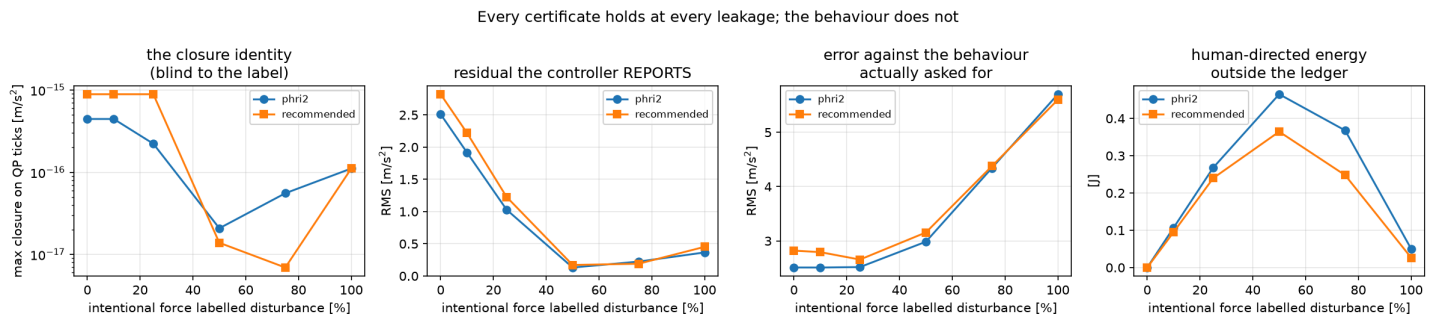
The pattern in the failures is §10’s anti-correlation appearing *within a single cell*: the runs with the largest  $r_{\text{auth}}$  (122–403 %) and the largest excursions (137–163 %) are exactly the runs where the precondition breaks. Loading both axes harder does not get you a better exhibit; it gets you a broken anchor.

**So (b) is not a one-off fluke — it recurs at three distinct seeds — but it is a minority outcome (25 %) that always sits outside the nominal box (111–120 % even when it qualifies).** It is not something to build a paper’s central claim on. §12’s decision 3 resolves to **(a), with (b) demoted to a remark**: that a dual-axis regime exists at all is worth one sentence and a pointer to the data; it is not the exhibit.

## 11. What this does not show

### 11.0 The force-misclassification pillar, measured

This sat on the owed list as “explicitly disclaim it”. A disclaimer that says *this could fail* is worth much less than one that says **how, and by how much**, so `pir_misclassification.py` sweeps it instead, using `impedance_residual`’s own knob: a fraction  $\lambda$  of the intentional force  $F_h$  is labelled disturbance, so the behaviour layer stops responding to it while the plant still feels all of it. **Only the label moves** — the total force on the plant is identical at every  $\lambda$ , so every difference is attributable to the labelling alone.



**Figure 12 — the pillar.** Panels 2 and 3 are the result: the residual the controller *reports* falls while the error against the behaviour actually asked for rises. The two poses almost coincide, which is worth noting on its own after §11.1’s table — this is the one finding here that is **not** a property of one FR3 configuration.

| $\lambda$ | certificates | closure               | reported residual | true error   | unaccounted energy |
|-----------|--------------|-----------------------|-------------------|--------------|--------------------|
| 0 %       | all hold     | $4.4 \times 10^{-16}$ | 2.510             | 2.510        | 0.000 J            |
| 10 %      | all hold     | $4.4 \times 10^{-16}$ | 1.915             | 2.510        | 0.107 J            |
| 25 %      | all hold     | $2.2 \times 10^{-16}$ | 1.029             | 2.517        | 0.268 J            |
| 50 %      | all hold     | $2.1 \times 10^{-17}$ | <b>0.131</b>      | 2.985        | 0.465 J            |
| 75 %      | all hold     | $5.6 \times 10^{-17}$ | 0.223             | 4.342        | 0.368 J            |
| 100 %     | all hold     | $1.1 \times 10^{-16}$ | 0.365             | <b>5.701</b> | 0.050 J            |

(at `phr12`’s pose; the recommended pose differs by under 10 % on every row.)

Three separate failures, and they are worth separating because only the first is what “the estimator is wrong” usually means.

**The certificates do not notice, and could not.** Merged Lemma 1 bounds torque, the tank bounds the residual’s port power, §8.6’s propositions bound feasibility. None of them is a statement about whether the label is right, so a mislabelled force violates none of them. Every row above is green.

**The best diagnostic in this document is structurally blind to it.** The four-term closure caught two real bugs (§8.2, §9.4) precisely because it is an identity rather than a fit. But it is evaluated against  $a_{id}$  computed from the *behaviour channel* — so when the label is wrong it closes to machine precision **against the wrong target**. An identity certifies consistency with what you told it to aim at, and nothing more. That is the sharpest single statement available about the pillar.

**The headline metric moves the wrong way.** This is the part I did not anticipate. The reported realization residual *falls* — 2.510 to 0.131, a factor of 19 — while the true behavioural error *rises*. The mechanism is plain once seen: with the push relabelled, the QP’s job becomes cancelling it, and the leaked  $a_{id}$  is easy to realize. So misclassification does not merely evade detection; **it registers as an improvement on the primary number this document reports throughout §7–§10**. There is no experiment in those sections that would distinguish “the realization got better” from “the labelling got worse”.

Two cautions on reading the table. The unaccounted-energy column is the energy exchanged through the *misclassified portion of  $F_h$  specifically*; it is not a claim that the ledger otherwise accounts for human power. It does not, and was never meant to — impedance\_residual’s tank is about the residual’s port power by design. And that column is **not monotone**: it peaks at  $\lambda = 0.5$  and collapses to 0.050 J at  $\lambda = 1$ , because a robot that ignores the push entirely behaves like a position controller — maximum behavioural error, minimum motion, minimum energy exchange. So it is not a detector either. The only monotone column is the one that requires knowing the true labels, which is exactly what is unavailable at run time.

**What follows for the write-up.** Not a caveat at the end. Every realization residual reported in §7–§10 is a number that misclassification pushes *downward*, and the paper has to say so where those numbers appear, not in a limitations paragraph a reader reaches after believing them. **No green cell in this document implies anything about this pillar**, and the architecture’s honest claim is narrower than it first looks: PIR certifies that the *declared* behaviour was realized within stated bounds. Whether the declaration matches what the human is doing is upstream of everything here.

11.1 A pattern in what turned out to be wrong

Ten conclusions in this document were later overturned or narrowed by a subsequent experiment — including one *correction* that was itself over-corrected — and the pattern is worth stating because it bears on how much anything here should be trusted:

| conclusion                                                       | fate                                                                                                                                                         |
|------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\rho = 0.28$ is <i>structurally</i> infeasible (§5.1)           | X pose-specific (§9.2)                                                                                                                                       |
| the $E_0$ trade is <i>flatter</i> , not steeper (§7.5)           | X pose artifact (§9.4)                                                                                                                                       |
| neither source benchmark stresses both axes (§7.4)               | X pose artifact (§9.4)                                                                                                                                       |
| the nominal dominates the command, 84 % (§7.6)                   | X pose artifact, → 46 % (§9.4)                                                                                                                               |
| the split makes a hard guarantee conditional (§7.7)              | ✓ true, but fixable (§8.2) and inert at the good pose (§8.4)                                                                                                 |
| anisotropic gains do not help <i>and slightly hurt</i> (§8.3)    | ~ the “hurt” is pose-specific; useless but harmless at the good pose (§8.4)                                                                                  |
| passivity peaks at intermediate $K_0$ (§10.2, first draft)       | X inverts at the good pose — committed <i>while writing this table</i>                                                                                       |
| $\alpha_{nom}$ is “inert while the anchor fits” (§8.2)           | ~ under the <i>monotone</i> rule, only from a fresh state: its latch broke inertness on 1326 of 4000 ticks (§8.5, C5)                                        |
| decision 0 “costs nothing measurable” (§12, as adopted)          | X §8.5’s (C4): 71 points of excursion for 0.74 % of a derated cap                                                                                            |
| that cost is the price of $\alpha_{nom}$ (§8.5, C4, first draft) | X it was the price of the <b>monotone rule</b> . Metering the re-stiffening against the tank instead returns all 71 points at 8× and most of the rest at 12× |

Five of ten were properties of **one FR3 configuration**, not of the architecture; three more (the last three rows) were claims about the *fix* that its own checker falsified — and the last of those is the sharpest case in the table, because it is a correction that went

too far: having found a real cost, this document attributed it to the fix as a whole rather than to the particular rule chosen to implement it, and only measuring a second rule separated them. In each case the erroneous generalisation was made from a carefully measured, internally consistent experiment. The measurements were right; the scope claimed for them was not.

**Nothing in this line should be claimed without a pose sweep** — a rule I restated in §10.2 and then broke in the next paragraph, which is the most honest evidence available for how strong the pull toward the tidy generalisation is. The two bugs that the four-term residual’s closure check caught (§8.2, §9.4) suggest the same about any diagnostic that cannot be checked against an identity.

- ~~The merged controller is not implemented.~~ Discharged in §7.2: the replay estimate and the merged closed loop agree to 0.1 % on both rows, and a regression test now pins them together.
- ~~The anchor’s feasibility does not survive a larger scenario.~~ Fixed in §8.2 by `nominal_auth_energy`, which holds both guarantees to 16×. What is *not* fixed is the condition underneath it: everything still assumes  $\tau_{\text{base}}$  alone fits the envelope, which is a property of the pose and is not defended anywhere.
- **The fix’s cost is not fully characterised over long horizons.** `nominal_auth_energy` recovers the floor’s stiffness as the tank can pay for it, which removes the within-run ratchet the monotone rule had (§8.5, C5). But a 6 s benchmark does not show what repeated contact does to the tank over minutes: recovery is funded by the damping the arm actually dissipates, so a sequence of transients with little motion between them could in principle keep  $\alpha_{\text{nom}}$  down indefinitely. That regime is not tested here.
- **The  $K_0$  trade is priced on one disturbance profile.** §8.3’s workspace numbers at 12× come from `impedance_residual`’s own rejectable force scaled up, at one seed. That is a stress test, not a distribution.
- **The fallback leaves the workspace box in transient.** 62.2 mm peak against a 60 mm bound. §12.0 settles that `phri2`’s box is slack-relaxed by necessity, so this is not a constraint violation in its formulation, and the 33-cell region stands. What remains true and worth keeping is the weaker statement: the  $\alpha \rightarrow 0$  guarantee is “**settles** inside the box”, not “stays inside it”, and nothing here bounds the transient.
- **One pose, one push direction, isotropic gains.** The whole gate is evaluated at `phri2`’s nominal hold pose under a  $-z$  push, with  $K_0 = kI$ ,  $D_0 = dI$ . Joint 4 is the binding joint in every single cell, and it is binding largely because of an 18.96 N·m bias torque at that specific configuration. An anisotropic  $(K_0, D_0)$  — stiffer along the push axis, softer elsewhere — is the obvious first lever if the region needs widening, and the note explicitly defers it. A different pose could move the verdict in either direction.
- **Recursive feasibility is proved only for the frozen model.** §8.6’s Propositions 2 and 3 give a sound feasibility certificate (0 unsound ticks in 1800) and an invariant terminal set, and at the recommended pose both hold at 100 % of ticks out to 12× disturbance. Neither closes the gap that matters:  $\tau_{\text{base}}$ ,  $J_v$  and  $\Lambda^{-1}$  are re-frozen every solve from the nonlinear state, so recursion still assumes the re-frozen model lands in  $X_f$ . The certificate is also conservative — 28.3 % of ticks fail it at `phri2`’s pose under 12× where only 4.3 % of solves actually fail.
- **The force-misclassification pillar is not defended — now measured, §11.0.** Every certificate holds at every leakage from 0 to 100 %, the closure identity stays at machine precision against the wrong target, and the reported residual *falls* by a factor of 19 while the true behavioural error more than doubles. **No green cell in this document implies anything about that pillar**, and the headline metric actively rewards getting the labels wrong.
- **Double tightening is not measured.** `phri2`’s horizon-wide torque tightening and `impedance_residual`’s tank tightening compound in series. That curve is Task 3 and is expected to get steeper after the merge.
- Simulation only, translational only, affine memoryless behaviour class.

## 11.2 Applying the rule: which claims survive a pose sweep

§11.1’s rule — *nothing in this line should be claimed without a pose sweep* — was stated and then broken in the next paragraph. Rather than restate it a third time, `pir_pose_sweep.py` applies it: take the claims this document still makes, evaluate each at every pose in §10.5’s family, and report which are pose-invariant.

**One variable.** The gain rule is held fixed —  $K_0 = 380$  with  $D_0$  from each pose’s own task inertia — so only the pose moves. That is deliberately *not* the recommended operating point, whose  $K_0 = 520$  was chosen for its pose by §9.3; sweeping both at once would conflate “the pose changed” with “the design changed”, which is how five of §11.1’s nine reversals happened.

| $\lambda$                | 0.0            | 0.2    | 0.4    | 0.6    | 0.8           | 1.0    |
|--------------------------|----------------|--------|--------|--------|---------------|--------|
| diagnostic 3 fallback    | 55.8 mm        | 53.2   | 50.7   | 48.6   | 46.5          | 45.3   |
| anchor headroom, 1×      | <b>2.4 %</b>   | 7.9 %  | 22.8 % | 45.9 % | <b>64.2 %</b> | 49.5 % |
| anchor headroom, 12×     | <b>−13.5 %</b> | 0.3 %  | 21.3 % | 43.1 % | 51.4 %        | 40.5 % |
| Prop 2 certificate, 12×  | 71.7 %         | 89.0 % | 99.0 % | 100 %  | 100 %         | 100 %  |
| Prop 3 terminal set, 12× | 90.0 %         | 96.3 % | 100 %  | 100 %  | 100 %         | 100 %  |
| hard box solves, 12×     | 60.0 %         | 93.3 % | 100 %  | 100 %  | 100 %         | 100 %  |
| infeasible solves, 12×   | <b>12</b>      | 0      | 0      | 0      | 0             | 0      |

Sixteen claims, and the split is exactly half:

| pose-invariant (8)                     | pose-dependent (8)                    |
|----------------------------------------|---------------------------------------|
| Lemma 1 <b>conclusion</b> , 1× and 12× | Lemma 1 <b>precondition</b> , 12×     |
| tank floor, 12×                        | anchor headroom positive, 12×         |
| diagnostic 3: fallback inside the box  | anchor headroom above 10 %, 1×        |
| $\alpha_{\text{nom}}$ inert at 1×      | $\alpha_{\text{nom}}$ inert at 12×    |
| Lemma 1 precondition, 1×               | Prop 2 certificate at every tick, 12× |
| anchor headroom positive, 1×           | Prop 3 terminal set nonempty, 12×     |
| §11.0's misclassification blindness    | a hard box would still solve, 12×     |
|                                        | no infeasible solve, 12×              |

**The split is not arbitrary, and it has a one-line summary: the guarantees are pose-invariant, the hypotheses are not.** Every claim that moves is a statement about the **feasibility axis** — the precondition, the headroom, the certificate, the terminal set — and every one of them fails at the  $\text{phri2}$  end and recovers by  $\lambda \approx 0.4$ . Nothing about the *conclusions* moves: the torque envelope and the tank floor hold at every pose at every load tested. That is Appendix A.4's structural point — (P1) is a standing run-time hypothesis, not a design-time fact — demonstrated across a family rather than argued.

Three things this changes in how the rest should be read.

**The pose dependence is really a pose × load interaction.** Every failing cell is at 12×; at 1× all sixteen claims hold at all six poses. So “§8 is a diagnosis of a configuration” (§8.4) is right but understated: it is a diagnosis of a configuration *under load*. A nominal-load benchmark at any pose in this family would have found nothing.

**A binary claim can hide a gradient, and one of mine did.** “Anchor headroom positive at 1×” is pose-invariant — and useless, because it is 2.4 % at  $\lambda = 0$  and 64.2 % at  $\lambda = 0.8$ . Raising the threshold to 10 % flips it to pose-dependent. I had

predicted the torque-axis claims would move and they split instead; the claim that did not move only failed to move because it was too weak to. Where this document reports headroom, the number matters and the sign does not.

**The recommended pose is not the best pose in this family.** Headroom peaks at  $\lambda = 0.8$  (64.2 % at  $1\times$ ) rather than at  $\lambda = 1.0$  (49.5 %). §9.3 selected  $\lambda = 1$  against a different objective and at its own  $K_0$ , so this is not a contradiction — but it does mean the pose recommendation is not “as far from `phri2` as possible”, and nothing in §9 established that it was.

**And the one finding that does not move at all is the uncomfortable one.** §11.0’s misclassification failure is pose-invariant in an unusually strong sense: the leaked runs converge to nearly the same place at every pose — reported residual 0.104–0.134 against a clean 2.04–2.51, true error 2.99–3.03. Six poses, and the failure mode lands within 3 % of itself each time. After nine reversals that were properties of one FR3 configuration, the one result here that generalises cleanly is the one about what the certificates cannot see.

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## 12. Decision gate

The synthesis note’s §13.3 put the decision gate right after Task 1. The work ran well past that, because each result changed what the next question was; what follows is where it actually stops.

`common_region_nonempty == true` under `phri2`’s derated-joint-4 envelope, the merged controller is implemented, Merged Lemma 1 holds in closed loop and the four-term residual closes exactly. What is left is not a coding decision.

§8, §9 and §10 have each changed what is being decided. Finding 4 is fixed and adopted; findings 1–3 turned out to be largely the pose; and §10 has moved the open question from the design to the **framing**. §12.0 then closes the two items that had been parked as unanswerable from inside the simulation — both turned out to be answerable by reading the sources, and neither answer was the one the question implied. Decisions 0, 2 and 3 are settled; decision 1 is the human’s and is probably moot; what is left open is theory, not design.

### 12.0 The two specification questions, answered from the source

Two items sat on the owed list as things “not answerable from inside the simulation”: what the derated torque envelope *means*, and whether the 0.06 m workspace bound is hard or slack-relaxed. Both are answerable — not by running anything, but by reading what the source repositories say about the numbers this document inherited from them. Neither answer is the one the question implied.

#### The derated envelope is neither thermal nor safety-certified

Both envelopes this document uses are labelled by their own sources as artificial. On  $\rho = 0.28$  (`impedance/impedance_residual.md` §5.1):

*The 28 % envelope represents a deliberately derated continuous budget, chosen before the accepted run to keep the nominal torque feasible while preserving headroom pressure for the residual; **it is not a manufacturer continuous-duty specification**, and the absolute FR3 limits remain the MuJoCo safety backstop regardless.*

And on the derated-joint-4 envelope that decision 2 selected (`imp_reference/paper.md` §6.4):

*joint 4’s available budget deliberately derated from its nominal 87 Nm to 31.5 Nm. **This is an artificial actuator-budget stress test, not a claim about the FR3’s physical rating.***

So the question “thermal budget or safety-certified limit?” has a third answer: **neither**.  $\rho \tau_{\max}$  is a stress budget chosen to make the torque constraint bind, with the real limit enforced separately underneath it. Three consequences.

First, **§8.5’s exchange rate is settled, not merely quantified**. The 0.74 % overrun the ablation incurs is an overrun of an artificial budget, at 36 % of the hardware limit. It is not a safety event, and it never was one. Decision 0 therefore does not turn on a specification this document was waiting for.

Second, and less comfortably, **the feasibility axis is measured entirely against a budget its own source calls a stress test**. That does not invalidate it — the architecture behaves the same way against any envelope tighter than the hardware, and genuinely tight envelopes exist (thermal derating, and the reduced-power modes a safety-rated pHRI deployment runs in). But it

does mean the *numbers* — 33 cells, 2.4 % headroom,  $\rho = 0.28$  — are properties of a chosen stress budget and not of the FR3, and the paper must say so where it reports them.

Third, it sharpens what §8.4 already found. At the recommended pose the torque axis never binds at any  $K_0$ . Combined with the above: the feasibility axis binds only at a pose chosen to load joint 4, against an envelope chosen to be tight. Both halves of that are deliberate, and both should be stated as deliberate.

### The workspace box is slack-relaxed, and cannot be made hard here

The box is soft in the implementation this document reuses, and the source says why. From

`imp_reference/simulation/fr3_interaction_dynamics_mpc.py` (the condensing routine, on the Cartesian position/speed rows):

*Cartesian workspace / speed box — **SOFT (slack-relaxed). Hard box constraints on predicted state have no recursive-feasibility guarantee** (paper.md Prop. 3): once the real, nonlinear MuJoCo trajectory drifts from this frozen-Jacobian prediction enough to sit right at an active bound, the very next solve can become genuinely infeasible (confirmed empirically — tightening or loosening the bound value alone did not remove it, nor did loosening the torque limit, isolating the box constraints as the cause).*

The penalty is  $w_s = 10^8$ , set after  $10^4$  “let the QP trade away the box far too readily (observed slack of several meters)”, and the source paper reports the resulting boundary held “to within approximately 0.1–0.2 mm”.

So the answer is not “hard” or “soft by choice”. **The box cannot be hard in this architecture** — making it hard was tried and breaks the QP, for exactly the reason §11 lists as still unproven (no recursive-feasibility guarantee). Anyone who wants a hard box has to supply that proof first; it is the same missing theorem in a different costume.

One further quotation settles the part of this that actually bit. The source’s *desired behaviour* violates the bound and the source is untroubled by it ( `imp_reference/paper.md` §6.3):

*The impedance reference has an unconstrained static displacement  $F_h/K_d = 20/200 = 0.10$  m, **which by itself already exceeds the 0.06 m bound**. ... Predictive realization instead departs from the desired behavioral acceleration while the bound is active, holding displacement to 0.0601 m.*

In `phri2` the box constrains the **realized trajectory**, softly, and the behaviour layer is explicitly allowed to want something outside it. Diagnostic 3 asks something different and stronger: that the **passive fallback alone** hold the push inside the same number, which forces  $K_0 \geq |F_h|/0.06 = 333$  N/m and is the single constraint §8.1 traced all three findings to.

That is a design choice this document is free to make — a bound on where the arm ends up if the authorization collapses is a reasonable thing to want — but it is **not inherited**, and this document has been citing it as though it were. Three corrections follow:

- **Task 1’s 33 cells stand; the 6-cell reading is withdrawn.** Gating on the fallback’s transient *peak* rather than its settled displacement tightens an already-stricter-than-source criterion against a box that is soft by necessity. The published region is the settled-displacement one.
- **soft\_nominal ( $K_0 = K_d = 200$ ) reads the premise correctly.** §8.3 put this as “rejects diagnostic 3’s premise rather than its arithmetic” and hedged; it is now quotable.  $K_d = 200$  is what the source itself runs, at a static displacement of 0.10 m, and it meets the bound through the predictive layer rather than the impedance. §8.3’s other objection to it — 110 mm of fallback displacement and 160 mm of excursion under load — is a real engineering objection and is unaffected.
- **§10.6’s candidate is admissible, and it changes nothing.** The one dual-axis, well-behaved cell sits 16 % over the box, which under the correct reading is a slack excursion rather than a violation. So one of the two objections to framing (b) is removed. The other — that it survives resampling in only 5 of 20 runs — is the one that actually decided decision 3, and it stands. **(b) stays demoted.** Recording this is the point of §11.1: an answer that goes the way you hoped is exactly when to check whether it changes the conclusion, and here it does not.

**Decision 0 — adopt nominal authorization? ADOPTED, re-opened by §8.5, and now closed again with a different rule.** The authorization quadruples the range over which both guarantees survive, and §8.4 showed it is inert at the recommended pose throughout — insurance rather than a load-bearing part. What changed twice is *how the re-stiffening is restrained*:

- **nominal\_auth\_mono , adopted first.** Forbidding the rise buys (C2) outright, but §8.5's (C5) showed  $\alpha_{\text{nom}}$  is then a **latch**: it never recovers within a run, so one transient permanently softens the nominal, and §8.5's (C4) measured the bill — 71 percentage points of workspace excursion at 8× for a 0.74 % overrun of the derated cap.
- **nominal\_auth\_energy , adopted now.** Metering the rise against the tank instead of forbidding it —  $\alpha_{\text{nom}}$  may rise by at most what  $E - E_{\text{min}}$  buys at the current spring energy — keeps (C1) and (C2) exactly (0 violations on 4000 adversarial ticks, worst  $E = E_{\text{min}}$ ), holds both guarantees to 16×, and **returns all 71 of those points at 8×** (65.66 mm against the ablation's 65.64 mm) and most of the rest at 12×. It strictly dominates the monotone rule on every column of §8.3.

So the trade that re-opened this decision has largely evaporated, and §12.0 disposes of the specification question underneath it:  $\rho \tau_{\text{max}}$  is neither a thermal budget nor a safety-certified limit but a deliberately artificial stress budget, so the overrun the ablation incurs is not a safety event. Decision 0 is closed. What survives is a **disclosure** obligation, not a decision: the feasibility axis's numbers are properties of a chosen budget and must be reported as such.

### Decision 3 — how is the two-axis claim stated? SETTLED by §10.6: formulation (a), with (b) demoted to a remark.

Both candidates were resampled across 10 seeds × 2 disturbance profiles:

- **(a) The flip — survives 20/20.** The passivity axis goes from exactly zero at `phri2`'s pose to 40–51 % of the intended behaviour at the recommended one, and the feasibility axis from small-but-nonzero to exactly zero, on a joint-angle change that improves every behavioural metric. One correction §10.6 forces: at 1× disturbance the feasibility axis never reaches the load-bearing threshold at *either* pose, so the claim is “**which axis can carry load at all flips**”, not “the axes exchange the load”. Say the weaker thing.
- **(b) The dual-axis exhibit — does not survive as stated.** §10.5's cell was a lucky seed. Resampled, it is dual-axis in 15/20 runs but keeps all its certificates in **5/20**, always failing on Lemma 1's precondition, and even when it qualifies it sits 111–120 % outside the box. It recurs at three distinct seeds so it is not noise, but a 25 % outcome is not a central claim's evidence.

This also **retires the box question as a framing blocker**. It still matters for the Task 1 region size, but the exhibit it would have licensed does not survive resampling, so the framing no longer waits on it.

### Decision 1 — how much workspace containment under disturbance is the predictive layer's authority worth?

*Probably moot now.* This was the real form of findings 1–3:  $K_0 = K_d = 200$  buys 6.8× the anchor headroom and 2.7× the residual share, and pays with 160 mm of excursion at 12× disturbance against 79 mm. §9.3 gets 19× the headroom and 3.2× the residual share from the pose instead, and *improves* the excursion to 59 mm. Unless the application pins the pose, take the pose and leave  $K_0$  alone. Still worth a human's eye, because “the application pins the pose” is exactly the kind of constraint this study cannot see.

**Decision 2 — which envelope does the merged paper claim? SETTLED by §9:** `phri2`'s derated-joint-4 envelope, at a pose that does not load joint 4.  $\rho = 0.28$  is reachable (§9.2 corrects §5.1's over-claim) but leaves at best 13 % of the cap for the nominal, which is not enough for the split. For the record, the three options as they stood before §9 ran:

1. **Write the merge against `phri2`'s derated-joint-4 envelope.**  $(K_0, D_0) = (380, 29.1)$ , region of 33 cells, margins of 2–7 %. Honest, and the source benchmark actually runs there. Costs: the tank's numbers from `impedance_residual` were measured under  $\rho = 0.28$  and would have to be re-derived, and the margins are thin enough that §7's first bullet could consume them.
2. **Keep  $\rho = 0.28$  and abandon the passive-nominal split**, retreating to “dissipativity directly on the realized port, no nominal” — a different and harder design, and a separate planning decision. This looked *forced* when §5.1 read its result as structural; §9.2 withdrew that, so it was never forced — only expensive.
3. **Change the scenario** — a different nominal pose, a lower push magnitude, or a push direction that does not load joint 4 — and re-run this gate. Cheap to try (the scan is ~19 min on 4 cores) and it attacks the actual binding quantity, which is an 18.96 N·m bias torque at one configuration rather than anything about impedance.

Option 3 is what §9 ran, and it won: the pose relieves findings 1, 3 and 4 at once and improves containment rather than trading it. Option 1 is therefore the recommendation, *at the §9.3 pose rather than `phri2`'s*. Option 2 is not forced — the passive-nominal split is viable, so there is no reason to retreat to dissipativity-on-the-realized-port.

A tension that decision 3 has now dissolved rather than resolved: decision 2 recommends the §9.3 pose while §10.5's dual-axis cell lives at `phri2`'s, so formulation (b) would have had the paper recommending one operating point and demonstrating its

central claim at another. §10.6 demoted (b), so the question no longer arises — but it is recorded because it is the kind of thing that would have been easy to gloss.

## Owed before any claim

- ✓ ~~Decision 0: adopt nominal authorization~~ — adopted as `nominal_auth_energy`, the controller's default, with `pir_no_nominal_auth` and `nominal_auth_mono` kept as the ablations.
- ✓ ~~Decision 2: which envelope~~ — §9. Derated-joint-4, at the §9.3 pose.
- ✓ ~~Decision 3: the framing (§10.4)~~ — resolved to (a) by §10.6's resampling and unchanged by §12.0's box reading, which removed the other objection to (b) without rescuing it. **Human ratification still wanted:** this is the paper's central claim.
- **Human:** decision 1, which §9.3 suggests is moot unless the application pins the pose.
- ✓ ~~Root-cause the three Section 7 findings~~ — §8.1. One constraint, monotone in  $K_0$ .
- ✓ ~~Fix the precondition failure~~ — §8.2, `nominal_auth_energy`. Both guarantees to  $16\times$ .
- ✓ ~~Try anisotropic ( $K_0, D_0$ )~~ — §8.3. It does not help here and slightly hurts; the displacement is 8:1 push-axis dominated.
- ✓ ~~Re-run the gate at a pose that does not load joint 4~~ — §9.3. 481 cells against 33, and 45.7 % headroom against 2.4 %.
- ✓ ~~Re-run Tasks 2 and 3 at the §9.3 pose~~ — §9.4. Both axes are now exercised by `phri2`'s own benchmark, and §7.5's "flatter" conclusion is corrected: the trade is 7.7 %, as the synthesis note predicted.
- ✓ ~~Re-run §8's root-cause and fix comparison at the recommended pose too~~ — §8.4, from `pir_rootcause_pose_q2m13_q4m13.json` and `pir_fixes_pose_q2m13_q4m13.json`. All nine candidates score identically at the ceiling there and  $\alpha_{\text{nom}}$  never fires, which is the finding: §8 diagnoses a problem the pose largely removes. The numbers a paper quotes should still be the recommended pose's.
- **Search poses properly** rather than on a  $q_2, q_4, q_6$  grid screened by one scenario, and check whether the recommended pose is one a real task would accept. §11.2 narrows this but does not discharge it: it sweeps *claims* along one interpolation line between two known poses, which answers "which claims are pose-invariant" without answering "is this the right pose". §11.2's own finding that headroom peaks at  $\lambda = 0.8$  rather than at the recommended  $\lambda = 1$  is direct evidence that the line was never searched, only traversed.
- ✓ ~~Joint pose  $\times K_0$  scan looking for a well-behaved dual-axis operating point~~ — §10.5. 96 cells; one candidate, conditional on the box reading.
- ✓ ~~Promote §10.5's candidate to a proper case study — repeat it across seeds and disturbance profiles~~ — §10.6. It does not survive: 5/20 on all certificates, and the 16 % excursion was a mid-range draw of a 100–163 % spread. Demoted to a remark.
- ✓ ~~Restate Merged Lemma 1 to say what is guaranteed when the precondition fails but  $\alpha_{\text{nom}}$  is active~~ — §8.5, Merged Lemma 1'. Checked clause by clause on 4000 adversarial ticks. It also produced two results that were not being looked for: (P3) is what makes the energy half provable at all, and (C4)'s exchange rate first re-opened decision 0 and then, once a second rule was measured, turned out to be the latch's bill rather than the fix's.
- ✓ ~~Implement the merged controller and re-measure rows 1 and 4 in its own closed loop~~ — §7.1–7.2. Replay and closed loop agree to 0.1 %.
- ✓ ~~Task 3: re-sweep the authorization vs tracking curve ( $E_0$ ) with both tightenings active~~ — §7.4–7.5. The trade is **flatter**, not steeper, and §7.5 says why that is a bad sign rather than a good one.
- ✓ ~~Task 2, the proof: write merged Lemma 1 (per joint  $\alpha_r$  closed form,  $\alpha_r \cdot \alpha_E$  on segment lemma) and merged Proposition 1 at submittable granularity~~ — **Appendix A**, with `pir_theory_check.py` checking each statement. Lemma A's content turns out to be that the admissible set is an *interval anchored at zero*, which is what makes the two authorizations composable without a joint feasibility argument (Lemma B). (P1) is stated as the standing run-time hypothesis §7.7 demanded, and Appendix A.4 tabulates the three separate roles it plays. **Still owed:** a statement covering the model-mismatch step — Lemmas A and B are exact algebra at one tick, but §8.6's recursion is about the frozen model and re-freezing is not covered.
- ✓ ~~Decide what the controller does when the precondition fails~~ — §8.2. Folding the nominal into the authorization loop works, and the passivity cost it first appeared to carry is removed by authorizing the re-stiffening against the tank rather than forbidding it. The escalation route back into `certified-realizability` is still the answer for the residual condition §8.4 names ( $\tau_{\text{base}}$  itself), which no local fix defends.
- ✓ ~~Re-run the Task 1 gate at a pose/push direction that does not load joint 4, and check whether the headroom problem is FR3-pose-specific~~ — §9.3 for the gate, §11.2 for the claim-level sweep. The headroom problem is pose-specific *and* load-specific: every pose-dependent claim fails only at  $12\times$ , and only at the `phri2` end of the family.

- ✓ ~~Implement and test the per-contact reset~~ ~~nominal\_auth\_mono~~ ~~needs~~ — §8.2, §8.5. Solved without a reset, and better: metering the re-stiffening against the tank ( `nominal_auth_energy` ) removes the latch *and* returns most of (C4)'s cost, where a per-contact reset would have needed an episode boundary the controller has no way to detect. What is still owed is the long-horizon behaviour: nothing here characterises the ratchet over minutes of repeated contact.
- ✓ ~~State what the derated envelope means~~ — §12.0. Neither thermal/duty-cycle nor safety-certified: both sources label their envelope a deliberately artificial stress budget, with the FR3's absolute limits kept as a separate backstop. Decision 0 does not turn on it. What it *does* oblige is a disclosure — the feasibility axis's numbers are properties of a chosen budget, not of the FR3, and must be reported that way.
- ✓ ~~Resolve whether the 0.06 m bound is hard or slack-relaxed~~ — §12.0. Slack-relaxed, and not by preference: a hard box has no recursive-feasibility guarantee here and was confirmed empirically to make the QP infeasible. Task 1's 33 cells stand and the 6-cell reading is withdrawn. Diagnostic 3 is a stronger requirement than the source's own, which this document had been citing as inherited; that is now stated.
- ✓ ~~Prove what can be proved about recursive feasibility~~ — §8.6. Proposition 2 (the certificate, which turns out to be diagnostic row 1b) and Proposition 3 (the terminal ellipsoid, nonempty iff (P1) is strict at the fallback equilibrium). **Still owed:** the model-mismatch step. The frozen-model result does not survive re-freezing, and closing that is the same problem Task 2 has to state honestly.
- ✓ ~~Explicitly disclaim the force misclassification pillar in whatever is written~~ — §11.0, and measured rather than disclaimed. §7.1's channel split made the assumption visible; the sweep shows what it costs. **Still owed:** the disclaimer has to travel with the residual numbers in §7–§10, not sit in a limitations section, because misclassification pushes exactly those numbers the flattering way.

## 13. Reproducing

```

cd simulation
python3 pir_knot_scan.py                # ~19 min on 4 cores (Task 1 gate)
python3 pir_knot_scan.py --rescore      # re-derive verdicts/figures from stored maps
python3 pir_verify.py                  # ~3 min (gate verification)
python3 run_pir_closed_loop.py --scenario push # ~35 s (Section 7.2, 7.6)
python3 run_pir_closed_loop.py --scenario merged # ~35 s (Section 7.3, 7.4)
python3 pir_e0_sweep.py                # ~6 min (Section 7.4, 7.5)
python3 pir_rootcause.py               # ~4 min (Section 8.1)
python3 pir_fixes.py                   # ~5 min (Section 8.3)
python3 pir_pose_study.py              # ~4 min (Section 9.1, 9.2)
python3 pir_knot_scan.py --pose -1.30 -1.30 1.571 \
--tag pose_q2m13_q4m13                # ~19 min (Section 9.3)
POSE="--pose -1.30 -1.30 1.571 --k0 520.0 --d0 56.13 --tag pose_q2m13_q4m13"
python3 run_pir_closed_loop.py $POSE --scenario push # (Section 9.4)
python3 run_pir_closed_loop.py $POSE --scenario merged # (Section 9.4)
python3 pir_e0_sweep.py $POSE          # ~25 min (Section 9.4)
python3 pir_rootcause.py --pose -1.30 -1.30 1.571 --tag pose_q2m13_q4m13
python3 pir_fixes.py --pose -1.30 -1.30 1.571 --k0 520.0 --d0 56.13 \
--tag pose_q2m13_q4m13                # ~9 min (Section 8)
python3 pir_axis_tension.py            # seconds; reads results/ (Section 10)
python3 pir_joint_scan.py              # ~8 min (Section 10.5)
python3 pir_robustness.py              # ~10 min (Section 10.6)
python3 pir_lemma_check.py             # ~2 min (Section 8.5)
python3 pir_recursive_feasibility.py   # ~8 min (Section 8.6)
python3 pir_theory_check.py            # ~3 min (Appendix A)
python3 pir_misclassification.py       # ~8 min (Section 11.0)
python3 pir_pose_sweep.py              # ~22 min (Section 11.2)
python3 pir_pose_sweep.py --rescore    # re-derive the claim table only
python3 -m pytest test_pir_knot_scan.py test_pir_controller.py -q

```

MuJoCo mesh assets are gitignored repo-wide and must be fetched from MuJoCo Menagerie into

```

../simulation/models/franka_fr3/assets/ . Pinned dependency versions are in ../impedance/simulation/requirements.txt ;
mujoco==3.10.0 matters, since 3.13 renamed MjData.qM and fr3_mujoco.py reads it.

```

## Appendix A. Task 2: the merged statements

The synthesis note’s Task 2 asks for merged Lemma 1 — the per-joint  $\alpha_\tau$  closed form and the  $\alpha_\tau \cdot \alpha_E$  on-segment lemma — and merged Proposition 1, at submittable granularity, with the precondition stated as a **standing run-time hypothesis** rather than a design-time check.

The pieces were built in the order the experiments demanded them, so they are scattered: Merged Lemma 1’ is in §8.5, Propositions 2 and 3 in §8.6, the four-term identity in §7.3. This appendix collects the ones that are statements about the **algebra** rather than about one FR3 configuration, which is the part that transfers. `pir_theory_check.py` checks each; the numbers quoted are from it.

## A.1 Lemma A — the closed form, and why it is an interval

**Lemma A.** Let  $a$  be the anchor,  $r = J_v^\top F_r$ , and  $\bar{\tau}$  the per-joint envelope. If  $|a_l| \leq \bar{\tau}_l$  for every  $l$ , then

$$\mathcal{A}(a, r) := \{\alpha \in [0, 1] : |a + \alpha r| \leq \bar{\tau}\} = [0, \alpha_\tau], \quad \alpha_\tau = \min \left( 1, \min_{l: r_l \neq 0} \frac{\bar{\tau}_l - \text{sgn}(r_l) a_l}{|r_l|} \right).$$

*Proof.* Joint  $l$  constrains  $\alpha$  by  $-\bar{\tau}_l \leq a_l + \alpha r_l \leq \bar{\tau}_l$ . For  $r_l > 0$  the upper side binds, giving  $\alpha \leq (\bar{\tau}_l - a_l)/r_l$ ; for  $r_l < 0$  the lower side binds, giving  $\alpha \leq (-\bar{\tau}_l - a_l)/r_l = (\bar{\tau}_l + a_l)/|r_l|$ . Both are  $(\bar{\tau}_l - \text{sgn}(r_l) a_l)/|r_l|$ . Rows with  $r_l = 0$  reduce to  $|a_l| \leq \bar{\tau}_l$ , the hypothesis. Each row’s admissible set is an interval containing 0 — again by the hypothesis — so the intersection with  $[0, 1]$  is  $[0, \alpha_\tau]$ .  $\square$

The content is not the formula, which is elementary; it is that  **$\mathcal{A}$  is an interval anchored at zero**. That is the only property Lemma B needs, and it is what makes two independent authorizations composable without a joint feasibility argument.

The hypothesis is Merged Lemma 1’s precondition. **When it fails,  $\mathcal{A}$  is empty** — not small, empty — and no scaling of the residual repairs it, because every  $\alpha$  leaves  $a_l$  where it is. That is §7.7’s finding stated as algebra, and it is the reason  $\alpha_{\text{nom}}$  has to exist: only moving the anchor itself can restore feasibility.

Checked on 20 000 random draws against the imported `torque_scale` (rows with  $r_l = 0$  deliberately included): **0 disagreements, worst gap 0.0**;  $\alpha_\tau$  admissible in every draw; and **0 failures of maximality** — nudging past  $\alpha_\tau$  always leaves the box, so the bound is tight rather than merely sufficient.

## A.2 Lemma B — composition on the segment

**Lemma B.** Under Lemma A’s hypothesis, every  $\beta \in [0, \alpha_\tau]$  satisfies  $|a + \beta r| \leq \bar{\tau}$ . In particular  $\alpha_E \alpha_\tau$  is admissible for any  $\alpha_E \in [0, 1]$ , so composing the two scalings — in either order — preserves the torque conclusion.

*Proof.* Immediate from Lemma A:  $\mathcal{A} = [0, \alpha_\tau]$  and  $\alpha_E \alpha_\tau$  lies in it.  $\square$

This is the entire content of “the two authorizations do not fight”, and it is worth saying why it is so cheap. The energy layer may only *shrink* the residual, and shrinking moves the command along the segment from  $a + \alpha_\tau r$  back toward the anchor  $a$ , which is interior. Had  $\alpha_E$  been allowed to grow the residual, or had the two layers scaled different quantities, no such argument would be available and the composition would need a joint feasibility proof.

Checked on 20 000 draws: 0 violations at interior  $\beta$ , 0 violations of the composed scale, **worst composed**  $|\tau|/\bar{\tau} = 1.000000$  — saturating the envelope exactly, never through it, which is the same signature §7.3 reports in closed loop.

## A.3 Merged Proposition 1

**Merged Proposition 1.** Under **(P1)**  $|\tau_{\text{base}}| \leq \bar{\tau}$ , **(P2)**  $E_0 \geq E_{\min}$  and **(P3)** the energy-authorized re-stiffening rule of §8.5:

**(i) Torque.**  $|\tau_\ell| \leq \bar{\tau}$  at every tick.

**(ii) Energy.** Over any interval, the energy the residual extracts at the port is bounded by the initial budget plus the damping actually applied:

$$\sum_\ell h F_{\text{applied}, \ell}^\top v_\ell \leq (E_0 - E_{\min}) + \sum_\ell h \alpha_{\text{nom}, \ell} v_\ell^\top D_0 v_\ell.$$

**(iii) Realization.** The realized behaviour differs from the desired one by exactly the four-term residual,  $a_{\text{modelled}} - a_{\text{id}} = r_{\text{reg}} + r_{\text{con}} + r_{\text{mod}} + r_{\text{auth}}$ .

- i. is Lemma B plus  $\alpha_{\text{nom}}$ , which supplies the anchor feasibility Lemma A assumes whenever  $\tau_{\text{base}}$  alone fits. (ii) is the storage-function form of §8.5's (C2), stated so that it does not depend on the tank's upper cap — capping discards credit, which only strengthens the bound. (iii) is an identity, not a fit; §7.3 measures its closure at  $4.4 \times 10^{-16}$ , and it has caught two real bugs.
- ii. is checked as an *accumulation* on closed loops rather than on sampled ticks, because a per-tick check cannot fail the way a long run can:

| pose        | disturbance | extracted | budget + dissipated | slack  |
|-------------|-------------|-----------|---------------------|--------|
| phri2       | 1×          | −0.19 J   | 0.73 J              | 0.92 J |
| phri2       | 4×          | 0.05 J    | 0.87 J              | 0.82 J |
| phri2       | 12×         | −0.43 J   | 2.50 J              | 2.93 J |
| recommended | 1×          | 0.44 J    | 0.92 J              | 0.48 J |
| recommended | 4×          | 0.39 J    | 1.04 J              | 0.65 J |
| recommended | 12×         | −0.16 J   | 1.18 J              | 1.34 J |

Two things to read from the slack column. A negative extraction means the residual *fed* the port over the run, which is why (ii) is stated as a bound on the net rather than as a per-tick sign condition. And the slack **grows** with disturbance rather than shrinking — the damping term on the right scales with  $\|v\|^2$ , so a harder push funds more authority than it spends. The bound is therefore loose exactly where one would expect it to bind, and (ii) on its own is **not** what keeps the tank off its floor. That is (C2)'s per-tick argument in §8.5, which the  $\alpha_E$  clamp enforces tick by tick; (ii) is the interval-level consequence, and the two should not be confused for one another.

### A.4 The precondition is a standing hypothesis, and it is load-bearing three times

Everything above is conditional on **(P1)**, and the write-up has to say so in a way that survives being skimmed. (P1) is a property of the pose and the trajectory. **Nothing in the controller defends it.** It is re-evaluated at every tick and it can fail at run time — §10.6 found it holding in 10 of 20 resampled runs at one cell, while the *conclusion* held in all 20, which is the gap that forced Merged Lemma 1' in the first place.

It is not one hypothesis used once. It appears in three distinct roles:

| where                          | what it buys                                           | what its failure costs                           |
|--------------------------------|--------------------------------------------------------|--------------------------------------------------|
| Merged Lemma 1''s precondition | $\mathcal{A} \neq \emptyset$ , so $\alpha_\tau$ exists | no scaling of $F_r$ can hold the envelope (§7.7) |
| §8.6's terminal set            | $c^* > 0$ , so $X_f$ is nonempty                       | Proposition 3 has no terminal set at all         |
| §8.6's feasibility certificate | the QP's own $F_r = 0$ fallback is admissible          | the certified fallback is not available          |

A paper that states (P1) once, early, as a design-time check would be misrepresenting all three. The honest framing is the one §12.0 arrived at from the other direction: **this is a document about what can be certified and when**, and the certificate's hypothesis is a run-time property that the architecture reports rather than guarantees. §8.4 measured where it sits — 0.636 of the cap at phri2 's pose, 0.433 at the recommended one — and §8.6 measured what happens when it slips: at phri2 's pose under 12× disturbance, 4.3 % of solves are genuinely infeasible.

**What Task 2 still does not have.** A statement covering the model-mismatch step. Lemmas A and B are exact algebra at one tick and transfer unchanged; Merged Proposition 1 (i)–(ii) hold per tick and accumulate; but Proposition 3's recursion is about the QP's frozen model, and re-freezing at the next solve is not covered by anything here. That is the one genuinely open theoretical item, and §11 records it as such.